

SOLIDWORKS API (.NET VBA C#) — Reference (117/337): IFeatureManager (part 4 of 9)

SOLIDWORKS API Documentation (converted from CHM)

IFeatureManager (part 4 of 9) (Type)

SOLIDWORKS API Help

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GetFeatures Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetFeatures Method (IFeatureManager)

Gets the features in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatures( _  
    ByVal ToplevelOnly As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim ToplevelOnly As System.Boolean  
Dim value As System.Object  
  
value = instance.GetFeatures(ToplevelOnly)
```

C#

```
System.object GetFeatures(  
    System.bool ToplevelOnly  
)
```

C++/CLI

```
System.Object^ GetFeatures(  
    & System.bool ToplevelOnly  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToplevelOnly

True to get only the features at the top level of the FeatureManager design tree, false to get the top-level features and all child features in the FeatureManager design tree

Return Value

Array of all of the features

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetFeatures](#).

Example

[Create Loft Body \(VB.NET\)](#)
[Create Loft Body \(VBA\)](#)
[Create Loft Body \(C#\)](#)

Remarks

If ToplevelOnly is false, then this method gets all of the feature and child features in this document. It does not get features in components.

The features that are returned by this method can be in any order. You should not rely on the order to indicate anything about children or parents. If hierarchy and order information is needed, then use [IPartDoc::FirstFeature](#) or [IPartDoc::IFirstFeature](#) or [IModelDoc2::FirstFeature](#) or [IModelDoc2::IFirstFeature](#) or [IFeature::GetNextFeature](#) or [IFeature::IGetNextFeature](#), [IFeature::GetFirstSubFeature](#) or [IFeature::IGetFirstSubFeature](#), and [IFeature::GetNextSubFeature](#) or [IFeature::IGetNextSubFeature](#) to retrieve your information.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::IGetFeatures Method](#)
[IFeatureManager::GetFeatureCount Method](#)

[IFeature::GetChildren Method](#)
[IFeature::GetOwnerFeature Method](#)
[IFeature::GetParents Method](#)
[IFeature::IGetChildCount Method](#)
[IFeature::IGetChildren Method](#)
[IFeature::IGetParentCount Method](#)
[IFeature::IGetParents Method](#)
[IComponent2::FirstFeature Method](#)

 Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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GetFeatureTreeRootItem Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetFeatureTreeRootItem Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::GetFeatureTreeRootItem2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureTreeRootItem() As TreeControlItem
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As TreeControlItem  
  
value = instance.GetFeatureTreeRootItem()
```

C#

```
TreeControlItem GetFeatureTreeRootItem()
```

C++/CLI

```
TreeControlItem^ GetFeatureTreeRootItem();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Root item in the FeatureManager design tree](#)

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetFeatureTreeRootItem](#).

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

 Availability

SOLIDWORKS 2008 SP2, Revision Number 16.2

SOLIDWORKS API Help

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GetFeatureTreeRootItem2 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetFeatureTreeRootItem2 Method (IFeatureManager)

Gets the root item of the FeatureManager design tree in the specified pane.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureTreeRootItem2( _  
    ByVal WhichPane As System.Integer _
```

) As TreeControlItem

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim WhichPane As System.Integer  
Dim value As TreeControlItem  
  
value = instance.GetFeatureTreeRootItem2(WhichPane)
```

C#

```
TreeControlItem GetFeatureTreeRootItem2(  
    System.int WhichPane  
)
```

C++/CLI

```
TreeControlItem^ GetFeatureTreeRootItem2(  
    & System.int WhichPane  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichPane

FeatureManager design tree pane as defined in [swFeatMgrPane_e](#)

Return Value

[ITreeControlItem](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetFeatureTreeRootItem2](#).

Example

[Expand or Collapse FeatureManager Design Tree Nodes \(VBA\)](#)
[Expand or Collapse FeatureManager Design Tree Nodes \(VB.NET\)](#)
[Expand or Collapse FeatureManager Design Tree Nodes \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2012 SP04, Revision 20.4

SOLIDWORKS API Help

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GetFlatPatternFolder Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetFlatPatternFolder Method (IFeatureManager)

Gets the interface to the flat-pattern folder feature in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFlatPatternFolder() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Object  
  
value = instance.GetFlatPatternFolder()
```

C#

```
System.Object GetFlatPatternFolder()
```

C++/CLI

```
System.Object^ GetFlatPatternFolder();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IFlatPatternFolder](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetFlatPatternFolder](#).

Example

[Get Flat-Pattern Folder Feature \(VBA\)](#)
[Get Flat-Pattern Folder Feature \(VB.NET\)](#)
[Get Flat-Pattern Folder Feature \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

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GetFreezeLocation Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetFreezeLocation Method (IFeatureManager)

Gets the location of the freeze bar in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFreezeLocation() As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim value As Feature  
  
value = instance.GetFreezeLocation()
```

C#

```
Feature GetFreezeLocation()
```

C++/CLI

```
Feature^ GetFreezeLocation();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IFeature](#); Null, if the freeze bar is at the top of the FeatureManager design tree

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetFreezeLocation](#).

Example

[Move Freeze Bar \(VBA\)](#)
[Move Freeze Bar \(VB.NET\)](#)
[Move Freeze Bar \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.EditFreeze Method](#)
[IModelDocExtension.NeedsRebuild2 Property](#)
[IModelDocExtension.UpdateFrozenFeatures Method](#)
[IFeature.HasFrozenUpdatePending Method](#)
[IFeature.IsFrozen Method](#)
[IFeature.MoveFreezeBarTo Method](#)
[IFeature.IsHiddenLock Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

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GetPreTrimmedBodies Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetPreTrimmedBodies Method (IFeatureManager)

Gets the temporary trimmed bodies using the specified target sheet (surface) body according to the trim tools previously defined by [IFeatureManager.PreTrimSurface](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPreTrimmedBodies(  
    ByVal TargetSurface As Body2  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim TargetSurface As Body2  
Dim value As System.Object  
  
value = instance.GetPreTrimmedBodies(TargetSurface)
```

C#

```
System.Object GetPreTrimmedBodies(  
    Body2 TargetSurface  
)
```

C++/CLI

```
System.Object^ GetPreTrimmedBodies(  
    & Body2^ TargetSurface  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TargetSurface
Target sheet [body](#)

Return Value

Array of temporary trimmed bodies

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.GetPreTrimmedBodies](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.PostTrimSurface Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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GetSelectionSetFolder Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetSelectionSetFolder Method (IFeatureManager)

Gets the Selection Sets folder.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSelectionSetFolder() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Object  
  
value = instance.GetSelectionSetFolder()
```

C#

```
System.object GetSelectionSetFolder()
```

C++/CLI

```
System.Object^ GetSelectionSetFolder();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Selection Sets folder](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetSelectionSetFolder](#).

Example

[Get Objects in Selection Set \(C#\)](#)
[Get Objects in Selection Set \(VB.NET\)](#)
[Get Objects in Selection Set \(VBA\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[SelectionSet Interface](#)
[SelectionSetItem Interface](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

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GetSheetMetalFolder Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetSheetMetalFolder Method (IFeatureManager)

Gets the interface to the sheet metal folder feature in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSheetMetalFolder() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Object  
  
value = instance.GetSheetMetalFolder()
```

C#

```
System.object GetSheetMetalFolder()
```

C++/CLI

```
System.Object^ GetSheetMetalFolder();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[ISheetMetalFolder](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetSheetMetalFolder](#).

Example

[Get Sheet Metal Folder Feature \(VBA\)](#)
[Get Sheet Metal Folder Feature \(VB.NET\)](#)
[Get Sheet Metal Folder Feature \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

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GetSlicingData Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetSlicingData Method (IFeatureManager)

Gets a new slicing data object with default parameter values.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSlicingData() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Object  
  
value = instance.GetSlicingData()
```

C#

```
System.object GetSlicingData()
```

C++/CLI

```
System.Object^ GetSlicingData();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[ISlicingData](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetSlicingData](#).

Example

See the [ISlicingData](#) example.

Remarks

Before calling this method to access default slicing parameters, you must pre-select the first reference entity/entities for slicing.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[FeatureManager::InsertSlicing_Method\(\)](#)

Availability

SOLIDWORKS 2020 FCS, Revision Number 28.0

SOLIDWORKS API Help

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GetStructureSystemFolders Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GetStructureSystemFolders Method (IFeatureManager)

Gets the structure system folders.

.NET Syntax

Visual Basic (Declaration)

```
Function GetStructureSystemFolders() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Object  
  
value = instance.GetStructureSystemFolders()
```

C#

```
System.object GetStructureSystemFolders()
```

C++/CLI

```
System.Object^ GetStructureSystemFolders();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [IFeatures](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::GetStructureSystemFolders](#).

Remarks

The features returned are [IStructureSystemFolders](#)s. Use [IFeature::GetSpecificFeature2](#) after calling this method to get the IStructureSystemFolders.
See the [IStructureSystemFolder](#) Remarks.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

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GroupComponentInstances Property (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : GroupComponentInstances Property (IFeatureManager)

Gets or sets whether to group the same components in the same configuration in an assembly into a folder in the FeatureManager design tree.

.NET Syntax

Visual Basic (Declaration)

```
Property GroupComponentInstances As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As System.Boolean

instance.GroupComponentInstances = value
value = instance.GroupComponentInstances
```

C#

```
System.bool GroupComponentInstances {get; set;}
```

C++/CLI

```
property System.bool GroupComponentInstances {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to group the same components in the same configuration in an assembly into a folder in the FeatureManager design tree, false to ungroup them

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.GroupComponentInstances](#).

Example

[Group and Ungroup Components \(C#\)](#)
[Group and Ungroup Components \(VB.NET\)](#)
[Group and Ungroup Components \(VBA\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IAssemblyDoc::UngroupComponents Method \(\)](#)
[IAssemblyDoc::ReorderComponents Method \(\)](#)
[IAssemblyDoc::ReorganizeComponents Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

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HideBodies Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : HideBodies Method (IFeatureManager)

Hides both solid and surface bodies in the model.

.NET Syntax

Visual Basic (Declaration)

```
Sub HideBodies()
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
instance.HideBodies()
```

C#

```
void HideBodies()
```

C++/CLI

```
void HideBodies();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.HideBodies](#).

Example

[Create Replace Face Feature \(VBA\)](#)
[Create Replace Face Feature \(VB.NET\)](#)
[Create Replace Face Feature \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.ShowBodies Method](#)
[IBody2::DisableDisplay_Property](#)
[HideShowBodies Method](#)
[IModelDoc2::HideSolidBody_Method](#)
[IModelDoc2::HideSolidBody_Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HideComponentSingleConfigurationOrDisplayStateNames Property (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : HideComponentSingleConfigurationOrDisplayStateNames Property (IFeatureManager)

Gets or sets whether to hide a component's only configuration or display state.

.NET Syntax

Visual Basic (Declaration)

```
Property HideComponentSingleConfigurationOrDisplayStateNames As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim value As System.Boolean  
  
instance.HideComponentSingleConfigurationOrDisplayStateNames = value  
  
value = instance.HideComponentSingleConfigurationOrDisplayStateNames
```

C#

```
System.bool HideComponentSingleConfigurationOrDisplayStateNames {get; set;}
```

C++/CLI

```
property System.bool HideComponentSingleConfigurationOrDisplayStateNames {  
    System.bool get();  
    void set ( &    System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to hide the single configuration or display state, false to display it

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::HideComponentSingleConfigurationOrDisplayStateNames](#).

Example

See the [IFeatureManager::SetComponentIdentifiers](#) example.

Remarks

This property:

- Works in both SOLIDWORKS Design and [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#).
- Is analogous to the **Do not show Configuration or Display State name if only one exists** check box on the Component Name and Description dialog that appears after right-clicking on the top-level component in the FeatureManager design tree and selecting **Tree Display > Component Name and Description**.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2022 SP01, Revision Number 30.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

HoleWizard Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : HoleWizard Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.HoleWizard2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function HoleWizard(  
    ByVal GenericHoleType As System.Integer, _  
    ByVal StandardIndex As System.Integer, _  
    ByVal FastenerTypeIndex As System.Integer, _  
    ByVal SSize As System.String, _  
    ByVal EndType As System.Short, _  
    ByVal Diameter As System.Double, _  
    ByVal Depth As System.Double, _  
    ByVal Value1 As System.Double, _  
    ByVal Value2 As System.Double, _  
    ByVal Value3 As System.Double, _  
    ByVal Value4 As System.Double, _  
    ByVal Value5 As System.Double, _  
    ByVal Value6 As System.Double, _  
    ByVal Value7 As System.Double, _  
    ByVal Value8 As System.Double, _  
    ByVal Value9 As System.Double, _  
    ByVal Value10 As System.Double, _  
    ByVal Value11 As System.Double, _
```

```
ByVal Value12 As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim GenericHoleType As System.Integer  
Dim StandardIndex As System.Integer  
Dim FastenerTypeIndex As System.Integer  
Dim SSize As System.String  
Dim EndType As System.Short  
Dim Diameter As System.Double  
Dim Depth As System.Double  
Dim Value1 As System.Double  
Dim Value2 As System.Double  
Dim Value3 As System.Double  
Dim Value4 As System.Double  
Dim Value5 As System.Double  
Dim Value6 As System.Double  
Dim Value7 As System.Double  
Dim Value8 As System.Double  
Dim Value9 As System.Double  
Dim Value10 As System.Double  
Dim Value11 As System.Double  
Dim Value12 As System.Double  
Dim value As Feature
```

```
value = instance.HoleWizard(GenericHoleType, StandardIndex, FastenerTypeIndex, SSize, EndType,  
Diameter, Depth, Value1, Value2, Value3, Value4, Value5, Value6, Value7, Value8, Value9, Value10,  
Value11, Value12)
```

C#

```
Feature HoleWizard(  
    System.int GenericHoleType,  
    System.int StandardIndex,  
    System.int FastenerTypeIndex,  
    System.string SSize,  
    System.short EndType,  
    System.double Diameter,  
    System.double Depth,  
    System.double Value1,  
    System.double Value2,  
    System.double Value3,  
    System.double Value4,  
    System.double Value5,  
    System.double Value6,  
    System.double Value7,  
    System.double Value8,  
    System.double Value9,  
    System.double Value10,  
    System.double Value11,  
    System.double Value12  
)
```

C++/CLI

```
Feature^ HoleWizard(  
& System.int GenericHoleType,  
& System.int StandardIndex,  
& System.int FastenerTypeIndex,  
& System.String^ SSize,  
& System.short EndType,  
& System.double Diameter,  
& System.double Depth,  
& System.double Value1,  
& System.double Value2,  
& System.double Value3,  
& System.double Value4,  
& System.double Value5,  
& System.double Value6,  
& System.double Value7,  
& System.double Value8,
```

```
& System.double Value9,  
& System.double Value10,  
& System.double Value11,  
& System.double Value12  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GenericHoleType

StandardIndex

FastenerTypeIndex

SSize

EndType

Diameter

Depth

Value1

Value2

Value3

Value4

Value5

Value6

Value7

Value8

Value9

Value10

Value11

Value12

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.HoleWizard](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

SOLIDWORKS API Help

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HoleWizard2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : HoleWizard2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.HoleWizard3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function HoleWizard2( _  
    ByVal GenericHoleType As System.Integer, _  
    ByVal StandardIndex As System.Integer, _  
    ByVal FastenerTypeIndex As System.Integer, _  
    ByVal SSize As System.String, _  
    ByVal EndType As System.Short, _  
    ByVal Diameter As System.Double, _  
    ByVal Depth As System.Double, _  
    ByVal Value1 As System.Double, _  
    ByVal Value2 As System.Double, _  
    ByVal Value3 As System.Double, _  
    ByVal Value4 As System.Double, _  
    ByVal Value5 As System.Double, _  
    ByVal Value6 As System.Double, _  
    ByVal Value7 As System.Double, _  
    ByVal Value8 As System.Double, _
```

```

ByVal Value9 As System.Double, _
ByVal Value10 As System.Double, _
ByVal Value11 As System.Double, _
ByVal Value12 As System.Double, _
ByVal ThreadClass As System.String, _
ByVal RevDir As System.Boolean, _
ByVal UseFeatScope As System.Boolean, _
ByVal UseAutoSelect As System.Boolean, _
ByVal AssemblyFeatureScope As System.Boolean, _
ByVal AutoSelectComponents As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim GenericHoleType As System.Integer
Dim StandardIndex As System.Integer
Dim FastenerTypeIndex As System.Integer
Dim SSize As System.String
Dim EndType As System.Short
Dim Diameter As System.Double
Dim Depth As System.Double
Dim Value1 As System.Double
Dim Value2 As System.Double
Dim Value3 As System.Double
Dim Value4 As System.Double
Dim Value5 As System.Double
Dim Value6 As System.Double
Dim Value7 As System.Double
Dim Value8 As System.Double
Dim Value9 As System.Double
Dim Value10 As System.Double
Dim Value11 As System.Double
Dim Value12 As System.Double
Dim ThreadClass As System.String
Dim RevDir As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim value As Feature

```

```

value = instance.HoleWizard2(GenericHoleType, StandardIndex, FastenerTypeIndex, SSize, EndType,
Diameter, Depth, Value1, Value2, Value3, Value4, Value5, Value6, Value7, Value8, Value9, Value10,
Value11, Value12, ThreadClass, RevDir, UseFeatScope, UseAutoSelect, AssemblyFeatureScope,
AutoSelectComponents)

```

C#

```

Feature HoleWizard2(
    System.int GenericHoleType,
    System.int StandardIndex,
    System.int FastenerTypeIndex,
    System.string SSize,
    System.short EndType,
    System.double Diameter,
    System.double Depth,
    System.double Value1,
    System.double Value2,
    System.double Value3,
    System.double Value4,
    System.double Value5,
    System.double Value6,
    System.double Value7,
    System.double Value8,
    System.double Value9,
    System.double Value10,
    System.double Value11,
    System.double Value12,
    System.string ThreadClass,
    System.bool RevDir,
    System.bool UseFeatScope,

```

```

System.bool UseAutoSelect,
System.bool AssemblyFeatureScope,
System.bool AutoSelectComponents
)

```

C++/CLI

```

Feature^ HoleWizard2(
& System.int GenericHoleType,
& System.int StandardIndex,
& System.int FastenerTypeIndex,
& System.String^ SSize,
& System.short EndType,
& System.double Diameter,
& System.double Depth,
& System.double Value1,
& System.double Value2,
& System.double Value3,
& System.double Value4,
& System.double Value5,
& System.double Value6,
& System.double Value7,
& System.double Value8,
& System.double Value9,
& System.double Value10,
& System.double Value11,
& System.double Value12,
& System.String^ ThreadClass,
& System.bool RevDir,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GenericHoleType

Hole type as defined in [swWzdGeneralHoleTypes_e](#)

StandardIndex

Standard property as defined in [swWzdHoleStandards_e](#)

FastenerTypeIndex

Screw type as defined in [swWzdHoleStandardFastenerTypes_e](#)

SSize

Size of the hole

EndType

End type as defined in [swEndConditions_e](#)

Diameter

Diameter of the hole in meters

Depth

Depth of the hole in meters

Value1

Hole-related parameter

Value2

Hole-related parameter

Value3

Hole-related parameter

Value4

Hole-related parameter

Value5

Hole-related parameter

Value6

Hole-related parameter

Value7

Hole-related parameter

Value8

Hole-related parameter

Value9

Hole-related parameter

Value10

Hole-related parameter

Value11

Hole-related parameter

Value12

Hole-related parameter

ThreadClass

One of the following thread classes:

- 1B
- 2B
- 3B

Applies to ANSI inch standard only

RevDir

True to reverse the direction of the hole, false to not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects

AssemblyFeatureScope

True if the assembly feature only affects selected bodies in the assembly, false if the assembly feature affects all bodies in the assembly

AutoSelectComponents

True to auto-select all affected components, false to only use the selected components

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::HoleWizard2](#).

Remarks

To add a hole at multiple locations, preselect the sketch points and do not specify any selection marks.

Screw Fit - As defined in [swWzdHoleScrewClearanceTypes_e](#) and only used if user does not specify a hole diameter; default value is NORMAL_FIT

Drill Angle - defaults to 118 degrees if not specified

Input Metrics - Input angles in radians; input values in meters

Parameters for this method:

- The FastenerTypeIndex parameter has to match the standard and be valid for that hole type or an error occurs.
- The SSize parameter must be a valid size for fastener type or an error occurs.
- The Value1 through Value12 parameters are different for each type of hole. The possible values are as follows, organized by hole type. SOLIDWORKS ignores parameters set to -1.

Value1 through Value12 for different types of holes:

Counterbore Holes

Head clearance - Added to either the input counterbore depth or the counterbore depth in the standard.

Parameters for all counterbore holes:

1. CounterBore Diameter
2. CounterBore Depth
3. HeadClearance
4. Screw Fit
5. Drill Angle
6. Near Csink Diameter
7. Near Csink Angle
8. Bot Csink Diameter
9. Bot Csink Angle
10. Far Csink Diameter
11. Far Csink Angle
12. Offset

Countersink Holes

Head clearance type - Value from [swWzdHoleCounterSinkHeadClearanceTypes_e](#), which represents how to add the head clearance values to the hole.

Parameters for all countersink holes:

1. Near Csink Diameter
2. Near Csink Angle
3. Head Clearance
4. Screw Fit
5. Drill Angle
6. Far Csink Diameter
7. Far Csink Angle
8. Offset
9. Head Clearance Type

Regular Holes

Parameters for all holes:

1. Screw Fit
2. Drill Angle
3. Near Csink Diameter
4. Near Csink Angle
5. Far Csink Diameter
6. Far Csink Angle
7. Offset

Pipe Tapped Holes

Tap Drill Depth - Depth can be set up as a ratio from the standard thread depth or the input thread depth. If the user does specify a tap drill depth, then the calculation from the thread depth is replaced with the specified depth.

Cosmetic Thread Type - Defaults to swCosmeticThreadNone from [swWzdHoleCosmeticThreadTypes_e](#).

Near countersink angle - Can be calculated from standard data as a default.

Parameters for all pthap holes:

1. Thread Depth
2. Near Csink Diameter
3. Near Csink Angle
4. Far Csink Diameter
5. Far Csink Angle
6. Drill Angle
7. Cosmetic Thread Type
8. Offset

Tap Holes

Tap Drill Depth - If the user specifies a tap drill depth, then SOLIDWORKS always uses this depth. Tap drill depths can be calculated:

1. For helicoil holes: Tap drill depth is calculated based on a constant * tap drill diameter + allowance for tap type (bottoming or plug).
2. For tapped holes: Tap drill depth is calculated from the thread length + advance + allowance for tap type (bottoming or plug).

Thread Depth - If the user specifies thread depth, then SOLIDWORKS always uses this depth. Tap drill depths can be calculated:

1. For helicoil holes: Thread depth is calculated based on a constant (determined by hole type i.e. Insert = $1.0 * \text{Dia}$) + thread depth from the standard * thread major diameter.
2. For tapped holes: based on $2 * \text{thread diameter}$.

Cosmetic Thread Type - Specified as defined in swWzdHoleCosmeticThreadTypes_e.

Hcoil Tap Type - Defaults to swTapTypePlug as defined in [swWzdHoleHcoilTapTypes_e](#); used only for the helicoil standard holes.

Thread end condition - Defaults to swEndThreadTypeBLIND as defined in [swWzdHoleThreadEndCondition_e](#).

Parameters for all ptap holes:

1. Tap drill diameter
2. Tap drill depth
3. Thread Depth
4. Near Csink Diameter
5. Near Csink Angle
6. Far Csink Diameter
7. Far Csink Angle
8. Drill Angle
9. Cosmetic Thread Type
10. Thread End Condition
11. Helicoil Tap Type
12. Offset

Legacy Holes

For legacy types of holes StandardIndex, FastenerTypeIndex, and SSize are not relevant, and SOLIDWORKS ignores them.

The sequence of parameters 6 to 19 for different types of legacy holes is as follows:

Simple

1. Diameter
2. Depth
3. Sketch ID

Tapered

1. Minor Diameter
2. Depth
3. Sketch ID
4. Major Diameter

Counterbored

1. Diameter
2. Depth
3. Sketch ID
4. C-Bore Diameter
5. C-Bore Depth

Countersunk

1. Diameter
2. Depth
3. Sketch ID
4. C-Sink Angle
5. C-Sink Diameter

Counterdrilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Drill Diameter
5. C-Drill Depth
6. C-Drill Angle

Simple Drilled

1. Diameter
2. Depth
3. Sketch ID
4. Drill Angle

Tapered Drilled

1. Minor Diameter
2. Depth
3. Sketch ID
4. Major Diameter
5. Drill Angle

C-Bored Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Bore Diameter
5. C-Bore Depth
6. Drill Angle

C-Sunk Drilled

1. Diameter
2. Depth

3. Sketch ID
4. C-Sink Diameter
5. C-Sink Angle
6. Drill Angle

C-Drilled Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Drill Diameter
5. C-Drill Depth
6. Drill Angle
7. C-Drill Angle

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IWizardHoleFeatureData2 Interface](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HoleWizard3 Method (IFeatureManager)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IFeatureManager Interface](#) : HoleWizard3 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.HoleWizard4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function HoleWizard3( _
    ByVal GenericHoleType As System.Integer, _
    ByVal StandardIndex As System.Integer, _
    ByVal FastenerTypeIndex As System.Integer, _
    ByVal SSize As System.String, _
    ByVal EndType As System.Short, _
    ByVal Diameter As System.Double, _
    ByVal Depth As System.Double, _
    ByVal Value1 As System.Double, _
    ByVal Value2 As System.Double, _
    ByVal Value3 As System.Double, _
    ByVal Value4 As System.Double, _
    ByVal Value5 As System.Double, _
    ByVal Value6 As System.Double, _
    ByVal Value7 As System.Double, _
    ByVal Value8 As System.Double, _
    ByVal Value9 As System.Double, _
    ByVal Value10 As System.Double, _
    ByVal Value11 As System.Double, _
    ByVal Value12 As System.Double, _
    ByVal ThreadClass As System.String, _
    ByVal RevDir As System.Boolean, _
    ByVal UseFeatScope As System.Boolean, _
    ByVal UseAutoSelect As System.Boolean, _
    ByVal AssemblyFeatureScope As System.Boolean, _
    ByVal AutoSelectComponents As System.Boolean, _
    ByVal PropagateFeatureToParts As System.Boolean _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim GenericHoleType As System.Integer
Dim StandardIndex As System.Integer
Dim FastenerTypeIndex As System.Integer
Dim SSize As System.String
Dim EndType As System.Short
Dim Diameter As System.Double
Dim Depth As System.Double
Dim Value1 As System.Double
```



```

Dim Value2 As System.Double
Dim Value3 As System.Double
Dim Value4 As System.Double
Dim Value5 As System.Double
Dim Value6 As System.Double
Dim Value7 As System.Double
Dim Value8 As System.Double
Dim Value9 As System.Double
Dim Value10 As System.Double
Dim Value11 As System.Double
Dim Value12 As System.Double
Dim ThreadClass As System.String
Dim RevDir As System.Boolean
Dim UseFeatScope As System.Boolean
Dim UseAutoSelect As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim PropagateFeatureToParts As System.Boolean
Dim value As Feature

```

```

value = instance.HoleWizard3(GenericHoleType, StandardIndex, FastenerTypeIndex, SSize, EndType,
Diameter, Depth, Value1, Value2, Value3, Value4, Value5, Value6, Value7, Value8, Value9, Value10,
Value11, Value12, ThreadClass, RevDir, UseFeatScope, UseAutoSelect, AssemblyFeatureScope,
AutoSelectComponents, PropagateFeatureToParts)

```

C#

```

Feature HoleWizard3(
    System.int GenericHoleType,
    System.int StandardIndex,
    System.int FastenerTypeIndex,
    System.string SSize,
    System.short EndType,
    System.double Diameter,
    System.double Depth,
    System.double Value1,
    System.double Value2,
    System.double Value3,
    System.double Value4,
    System.double Value5,
    System.double Value6,
    System.double Value7,
    System.double Value8,
    System.double Value9,
    System.double Value10,
    System.double Value11,
    System.double Value12,
    System.string ThreadClass,
    System.bool RevDir,
    System.bool UseFeatScope,
    System.bool UseAutoSelect,
    System.bool AssemblyFeatureScope,
    System.bool AutoSelectComponents,
    System.bool PropagateFeatureToParts
)

```

C++/CLI

```

Feature^ HoleWizard3(
& System.int GenericHoleType,
& System.int StandardIndex,
& System.int FastenerTypeIndex,
& System.String^ SSize,
& System.short EndType,
& System.double Diameter,
& System.double Depth,
& System.double Value1,
& System.double Value2,
& System.double Value3,
& System.double Value4,
& System.double Value5,
& System.double Value6,
& System.double Value7,

```

```

& System.double Value8,
& System.double Value9,
& System.double Value10,
& System.double Value11,
& System.double Value12,
& System.String^ ThreadClass,
& System.bool RevDir,
& System.bool UseFeatScope,
& System.bool UseAutoSelect,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents,
& System.bool PropagateFeatureToParts
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GenericHoleType

Hole type as defined in [swWzdGeneralHoleTypes_e](#)

StandardIndex

Standard property as defined in [swWzdHoleStandards_e](#)

FastenerTypeIndex

Screw type as defined in [swWzdHoleStandardFastenerTypes_e](#)

SSize

Size of the hole

EndType

End type as defined in [swEndConditions_e](#)

Diameter

Diameter of the hole in meters

Depth

Depth of the hole in meters

Value1

Hole-related parameter

Value2

Hole-related parameter

Value3

Hole-related parameter

Value4

Hole-related parameter

Value5

Hole-related parameter

Value6

Hole-related parameter

Value7

Hole-related parameter

Value8

Hole-related parameter

Value9

Hole-related parameter

Value10

Hole-related parameter

Value11

Hole-related parameter

Value12

Hole-related parameter

ThreadClass

One of the following thread classes:

- 1B
- 2B
- 3B

Applies to ANSI inch standard only

RevDir

True to reverse the direction of the hole, false to not

UseFeatScope

True if the feature only affects selected bodies, false if the feature affects all bodies

UseAutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects

AssemblyFeatureScope

True if the assembly feature only affects selected bodies in the assembly, false if the assembly feature affects all bodies in the assembly

AutoSelectComponents

True to auto-select all affected components, false to only use the selected components

PropagateFeatureToParts

True to propagate the assembly feature to the components, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::HoleWizard3](#).

Remarks

To add a hole at multiple locations, preselect the sketch points and do not specify any selection marks.

Screw Fit - As defined in [swWzdHoleScrewClearanceTypes_e](#) and only used if user does not specify a hole diameter; default value is NORMAL_FIT

Drill Angle - defaults to 118 degrees if not specified

Input Metrics - Input angles in radians; input values in meters

Parameters for this method:

- The FastenerTypeIndex parameter has to match the standard and be valid for that hole type, or an error occurs.
- The SSize parameter must be a valid size for fastener type, or an error occurs.
- The Value1 through Value12 parameters are different for each type of hole. The possible values are as follows, organized by hole type. SOLIDWORKS ignores parameters set to -1.

Value1 through Value12 for different types of holes:

Counterbore Holes

Head clearance - Added to either the input counterbore depth or the counterbore depth in the standard.

Parameters for all counterbore holes:

1. CounterBore Diameter
2. CounterBore Depth
3. HeadClearance
4. Screw Fit
5. Drill Angle
6. Near Csink Diameter
7. Near Csink Angle
8. Bot Csink Diameter
9. Bot Csink Angle
10. Far Csink Diameter
11. Far Csink Angle
12. Offset

Countersink Holes

Head clearance type - Value from [swWzdHoleCounterSinkHeadClearanceTypes_e](#), which represents how to add the head clearance values to the hole.

Parameters for all countersink holes:

1. Near Csink Diameter
2. Near Csink Angle
3. Head Clearance
4. Screw Fit
5. Drill Angle
6. Far Csink Diameter
7. Far Csink Angle
8. Offset
9. Head Clearance Type

Regular Holes

Parameters for all holes:

1. Screw Fit
2. Drill Angle
3. Near Csink Diameter
4. Near Csink Angle
5. Far Csink Diameter
6. Far Csink Angle
7. Offset

Pipe Tapped Holes

Tap Drill Depth - Depth can be set up as a ratio from the standard thread depth or the input thread depth. If the user does specify a tap drill depth, then the calculation from the thread depth is replaced with the specified depth.

Cosmetic Thread Type - Defaults to swCosmeticThreadNone from [swWzdHoleCosmeticThreadTypes_e](#).

Near countersink angle - Can be calculated from standard data as a default.

Parameters for all ptap holes:

1. Thread Depth
2. Near Csink Diameter
3. Near Csink Angle
4. Far Csink Diameter
5. Far Csink Angle
6. Drill Angle
7. Cosmetic Thread Type
8. Offset

Tap Holes

Tap Drill Depth - If the user specifies a tap drill depth, then SOLIDWORKS always uses this depth. Tap drill depths can be calculated:

1. For helicoil holes: Tap drill depth is calculated based on a constant * tap drill diameter + allowance for tap type (bottoming or plug).
2. For tapped holes: Tap drill depth is calculated from the thread length + advance + allowance for tap type (bottoming or plug).

Thread Depth - If the user specifies thread depth, then SOLIDWORKS always uses this depth. Tap drill depths can be calculated:

1. For helicoil holes: Thread depth is calculated based on a constant (determined by hole type i.e. Insert = $1.0 * \text{Dia}$) + thread depth from the standard * thread major diameter.
2. For tapped holes: based on $2 * \text{thread diameter}$.

Cosmetic Thread Type - Specified as defined in swWzdHoleCosmeticThreadTypes_e.

Hcoil Tap Type - Defaults to swTapTypePlug as defined in [swWzdHoleHcoilTapTypes_e](#); used only for the helicoil standard holes.

Thread end condition - Defaults to swEndThreadTypeBLIND as defined in [swWzdHoleThreadEndCondition_e](#).

Parameters for all ptap holes:

1. Tap drill diameter
2. Tap drill depth
3. Thread Depth
4. Near Csink Diameter
5. Near Csink Angle
6. Far Csink Diameter
7. Far Csink Angle
8. Drill Angle
9. Cosmetic Thread Type
10. Thread End Condition
11. Helicoil Tap Type
12. Offset

Legacy Holes

For legacy types of holes StandardIndex, FastenerTypeIndex, and SSize are not relevant, and SOLIDWORKS ignores them.

The sequence of parameters 6 to 19 for different types of legacy holes is as follows:

Simple

1. Diameter
2. Depth
3. Sketch ID

Tapered

1. Minor Diameter
2. Depth
3. Sketch ID
4. Major Diameter

Counterbored

1. Diameter
2. Depth
3. Sketch ID
4. C-Bore Diameter
5. C-Bore Depth

Countersunk

1. Diameter
2. Depth
3. Sketch ID
4. C-Sink Angle
5. C-Sink Diameter

Counterdrilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Drill Diameter
5. C-Drill Depth
6. C-Drill Angle

Simple Drilled

1. Diameter
2. Depth
3. Sketch ID
4. Drill Angle

Tapered Drilled

1. Minor Diameter
2. Depth
3. Sketch ID
4. Major Diameter
5. Drill Angle

C-Bored Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Bore Diameter
5. C-Bore Depth
6. Drill Angle

C-Sunk Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Sink Diameter
5. C-Sink Angle
6. Drill Angle

C-Drilled Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Drill Diameter
5. C-Drill Depth
6. Drill Angle
7. C-Drill Angle

See Also

[FeatureManager Interface](#)
[FeatureManager Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

HoleWizard4 Method (IFeatureManager)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IFeatureManager Interface : HoleWizard4 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.HoleWizard5](#).

.NET Syntax

Visual Basic (Declaration)

```
Function HoleWizard4( _  
    ByVal GenericHoleType As System.Integer, _  
    ByVal StandardIndex As System.Integer, _  
    ByVal FastenerTypeIndex As System.Integer, _  
    ByVal SSize As System.String, _  
    ByVal EndType As System.Short, _  
    ByVal Diameter As System.Double, _  
    ByVal Depth As System.Double, _  
    ByVal Value1 As System.Double, _  
    ByVal Value2 As System.Double, _  
    ByVal Value3 As System.Double, _  
    ByVal Value4 As System.Double, _  
    ByVal Value5 As System.Double, _  
    ByVal Value6 As System.Double, _  
    ByVal Value7 As System.Double, _  
    ByVal Value8 As System.Double, _  
    ByVal Value9 As System.Double, _  
    ByVal Value10 As System.Double, _  
    ByVal Value11 As System.Double, _  
    ByVal Value12 As System.Double, _  
    ByVal ThreadClass As System.String, _  
    ByVal RevDir As System.Boolean, _  
    ByVal UseFeatScope As System.Boolean, _  
    ByVal UseAutoSelect As System.Boolean, _  
    ByVal AssemblyFeatureScope As System.Boolean, _  
    ByVal AutoSelectComponents As System.Boolean, _  
    ByVal PropagateFeatureToParts As System.Boolean _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim GenericHoleType As System.Integer  
Dim StandardIndex As System.Integer  
Dim FastenerTypeIndex As System.Integer  
Dim SSize As System.String  
Dim EndType As System.Short  
Dim Diameter As System.Double  
Dim Depth As System.Double  
Dim Value1 As System.Double  
Dim Value2 As System.Double  
Dim Value3 As System.Double  
Dim Value4 As System.Double  
Dim Value5 As System.Double  
Dim Value6 As System.Double  
Dim Value7 As System.Double  
Dim Value8 As System.Double  
Dim Value9 As System.Double  
Dim Value10 As System.Double  
Dim Value11 As System.Double  
Dim Value12 As System.Double  
Dim ThreadClass As System.String  
Dim RevDir As System.Boolean  
Dim UseFeatScope As System.Boolean  
Dim UseAutoSelect As System.Boolean  
Dim AssemblyFeatureScope As System.Boolean  
Dim AutoSelectComponents As System.Boolean  
Dim PropagateFeatureToParts As System.Boolean  
Dim value As Feature
```

```
value = instance.HoleWizard4(GenericHoleType, StandardIndex, FastenerTypeIndex, SSize, EndType,
```



Diameter, Depth, Value1, Value2, Value3, Value4, Value5, Value6, Value7, Value8, Value9, Value10, Value11, Value12, ThreadClass, RevDir, UseFeatScope, UseAutoSelect, AssemblyFeatureScope, AutoSelectComponents, PropagateFeatureToParts)

C#

```
Feature HoleWizard4(  
    System.int GenericHoleType,  
    System.int StandardIndex,  
    System.int FastenerTypeIndex,  
    System.string SSize,  
    System.short EndType,  
    System.double Diameter,  
    System.double Depth,  
    System.double Value1,  
    System.double Value2,  
    System.double Value3,  
    System.double Value4,  
    System.double Value5,  
    System.double Value6,  
    System.double Value7,  
    System.double Value8,  
    System.double Value9,  
    System.double Value10,  
    System.double Value11,  
    System.double Value12,  
    System.string ThreadClass,  
    System.bool RevDir,  
    System.bool UseFeatScope,  
    System.bool UseAutoSelect,  
    System.bool AssemblyFeatureScope,  
    System.bool AutoSelectComponents,  
    System.bool PropagateFeatureToParts  
)
```

C++/CLI

```
Feature^ HoleWizard4(  
& System.int GenericHoleType,  
& System.int StandardIndex,  
& System.int FastenerTypeIndex,  
& System.String^ SSize,  
& System.short EndType,  
& System.double Diameter,  
& System.double Depth,  
& System.double Value1,  
& System.double Value2,  
& System.double Value3,  
& System.double Value4,  
& System.double Value5,  
& System.double Value6,  
& System.double Value7,  
& System.double Value8,  
& System.double Value9,  
& System.double Value10,  
& System.double Value11,  
& System.double Value12,  
& System.String^ ThreadClass,  
& System.bool RevDir,  
& System.bool UseFeatScope,  
& System.bool UseAutoSelect,  
& System.bool AssemblyFeatureScope,  
& System.bool AutoSelectComponents,  
& System.bool PropagateFeatureToParts  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GenericHoleType
Hole type as defined in [swWzdGeneralHoleTypes_e](#)
StandardIndex
Standard property as defined in [swWzdHoleStandards_e](#)
FastenerTypeIndex
Screw type as defined in [swWzdHoleStandardFastenerTypes_e](#)
SSize

Size of the hole
EndType
End type as defined in [swEndConditions_e](#)
Diameter
Diameter of the hole in meters
Depth
Depth of the hole in meters
Value1
Hole-related parameter
Value2
Hole-related parameter
Value3
Hole-related parameter
Value4
Hole-related parameter
Value5
Hole-related parameter
Value6
Hole-related parameter
Value7
Hole-related parameter
Value8
Hole-related parameter
Value9
Hole-related parameter
Value10
Hole-related parameter
Value11
Hole-related parameter
Value12
Hole-related parameter
ThreadClass

One of the following thread classes:

- 1B
- 2B
- 3B

Applies to ANSI inch standard only

RevDir
True to reverse the direction of the hole, false to not (see **Remarks**)
UseFeatScope
True if the feature only affects selected bodies, false if the feature affects all bodies
UseAutoSelect
True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects
AssemblyFeatureScope
True if the assembly feature only affects selected bodies in the assembly, false if the assembly feature affects all bodies in the assembly
AutoSelectComponents
True to auto-select all affected components, false to only use the selected components
PropagateFeatureToParts
True to propagate the assembly feature to the components, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::HoleWizard4](#).

Remarks

To add a hole at multiple locations, preselect the sketch points and do not specify any selection marks.

Before creating a hole using a reference plane and IFeatureManager::HoleWizard4, determine the value of the RevDir parameter as follows:

- If the material is on one side of the reference plane, then RevDir is false.
- If the material is on both sides of the reference plane, then compare the distances between the reference plane and the material on both sides. If the distance between the reference plane and the material along its normal direction is bigger, then RevDir is true; otherwise, RevDir is false.

You can also use [IModelDoc2::RayIntersections](#) and [IModelDoc2::GetRayIntersectionsPoints](#) to find out if the material is on one side or both sides and to compare the distances.

The screw fit is defined as [swWzdHoleScrewClearanceTypes_e](#) and only used if a hole diameter is not specified; default value is NORMAL_FIT. The drill angle defaults to 118, if not specified. Input angles are in radians; input values are in meters.

Parameters for this method:

- The FastenerTypeIndex parameter has to match the standard and be valid for that hole type, or an error occurs.
- The SSize parameter must be a valid size for fastener type, or an error occurs.
- The Value1 through Value12 parameters are different for each type of hole. The possible values are as follows, organized by hole type. SOLIDWORKS ignores parameters set to -1.

Counterbore Holes

Head clearance - Added to either the input counterbore depth or the counterbore depth in the standard.

Parameters for all counterbore holes:

1. CounterBore Diameter
2. CounterBore Depth
3. HeadClearance
4. Screw Fit
5. Drill Angle
6. Near Csink Diameter
7. Near Csink Angle
8. Bot Csink Diameter

9. Bot Csink Angle
10. Far Csink Diameter
11. Far Csink Angle
12. Offset

Countersink Holes

Head clearance type - Value from [swWzdHoleCounterSinkHeadClearanceTypes_e](#), which represents how to add the head clearance values to the hole.

Parameters for all countersink holes:

1. Near Csink Diameter
2. Near Csink Angle
3. Head Clearance
4. Screw Fit
5. Drill Angle
6. Far Csink Diameter
7. Far Csink Angle
8. Offset
9. Head Clearance Type

Regular Holes

Parameters for all holes:

1. Screw Fit
2. Drill Angle
3. Near Csink Diameter
4. Near Csink Angle
5. Far Csink Diameter
6. Far Csink Angle
7. Offset

Pipe Tapped Holes

Tap Drill Depth - Depth can be set up as a ratio from the standard thread depth or the input thread depth. If the user does specify a tap drill depth, then the calculation from the thread depth is replaced with the specified depth.

Cosmetic Thread Type - Defaults to swCosmeticThreadNone from [swWzdHoleCosmeticThreadTypes_e](#).

Near countersink angle - Can be calculated from standard data as a default.

Parameters for all ptap holes:

1. Thread Depth
2. Near Csink Diameter
3. Near Csink Angle
4. Far Csink Diameter
5. Far Csink Angle
6. Drill Angle
7. Cosmetic Thread Type
8. Offset

Tap Holes

Tap Drill Depth - If the user specifies a tap drill depth, then SOLIDWORKS always uses this depth. Tap drill depths can be calculated:

1. For helicoil holes: Tap drill depth is calculated based on a constant * tap drill diameter + allowance for tap type (bottoming or plug).
2. For tapped holes: Tap drill depth is calculated from the thread length + advance + allowance for tap type (bottoming or plug).

Thread Depth If the user specifies thread depth, then SOLIDWORKS always uses this depth. Tap drill depths can be calculated:

1. For helicoil holes: Thread depth is calculated based on a constant (determined by hole type i.e. Insert = 1.0 * diameter) + thread depth from the standard * thread major diameter.
2. For tapped holes: based on 2 * thread diameter.

Cosmetic Thread Type - Specified as defined in swWzdHoleCosmeticThreadTypes_e.

Hcoil Tap Type - Defaults to swTapTypePlug as defined in [swWzdHoleHcoilTapTypes_e](#); used only for the helicoil standard holes.

Thread end condition - Defaults to swEndThreadTypeBLIND as defined in [swWzdHoleThreadEndCondition_e](#).

Parameters for all ptap holes:

1. Tap drill diameter
2. Tap drill depth
3. Thread Depth
4. Near Csink Diameter
5. Near Csink Angle
6. Far Csink Diameter
7. Far Csink Angle
8. Drill Angle
9. Cosmetic Thread Type
10. Thread End Condition
11. Helicoil Tap Type
12. Offset

Legacy Holes

For legacy types of holes StandardIndex, FastenerTypeIndex, and SSize are not relevant and SOLIDWORKS ignores them.

The sequence of parameters 6 to 19 for different types of legacy holes is as follows:

Simple

1. Diameter
2. Depth
3. Sketch ID

Tapered

1. Minor Diameter
2. Depth
3. Sketch ID

4. Major Diameter
- Counterbored**
1. Diameter
 2. Depth
 3. Sketch ID
 4. C-Bore Diameter
 5. C-Bore Depth

- Countersunk**
1. Diameter
 2. Depth
 3. Sketch ID
 4. C-Sink Angle
 5. C-Sink Diameter

- Counterdrilled**
1. Diameter
 2. Depth
 3. Sketch ID
 4. C-Drill Diameter
 5. C-Drill Depth
 6. C-Drill Angle

- Simple Drilled**
1. Diameter
 2. Depth
 3. Sketch ID
 4. Drill Angle

- Tapered Drilled**
1. Minor Diameter
 2. Depth
 3. Sketch ID
 4. Major Diameter
 5. Drill Angle

- C-Bored Drilled**
1. Diameter
 2. Depth
 3. Sketch ID
 4. C-Bore Diameter
 5. C-Bore Depth
 6. Drill Angle

- C-Sunk Drilled**
1. Diameter
 2. Depth
 3. Sketch ID
 4. C-Sink Diameter
 5. C-Sink Angle
 6. Drill Angle

- C-Drilled Drilled**
1. Diameter
 2. Depth
 3. Sketch ID
 4. C-Drill Diameter
 5. C-Drill Depth
 6. Drill Angle
 7. C-Drill Angle

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HoleWizard5 Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : HoleWizard5 Method (IFeatureManager)

Creates customized holes of various kinds.

.NET Syntax

Visual Basic (Declaration)

```
Function HoleWizard5( _  
    ByVal GenericHoleType As System.Integer, _  
    ByVal StandardIndex As System.Integer, _  
    ByVal FastenerTypeIndex As System.Integer, _  
    ByVal SSize As System.String, _  
    ByVal EndType As System.Short, _  
    ByVal Diameter As System.Double, _  
    ByVal Depth As System.Double, _  
    ByVal Length As System.Double, _
```

```

ByVal Value1 As System.Double, _
ByVal Value2 As System.Double, _
ByVal Value3 As System.Double, _
ByVal Value4 As System.Double, _
ByVal Value5 As System.Double, _
ByVal Value6 As System.Double, _
ByVal Value7 As System.Double, _
ByVal Value8 As System.Double, _
ByVal Value9 As System.Double, _
ByVal Value10 As System.Double, _
ByVal Value11 As System.Double, _
ByVal Value12 As System.Double, _
ByVal ThreadClass As System.String, _
ByVal RevDir As System.Boolean, _
ByVal FeatureScope As System.Boolean, _
ByVal AutoSelect As System.Boolean, _
ByVal AssemblyFeatureScope As System.Boolean, _
ByVal AutoSelectComponents As System.Boolean, _
ByVal PropagateFeatureToParts As System.Boolean _
) As Feature

```

Visual Basic (Usage)



```

Dim instance As IFeatureManager
Dim GenericHoleType As System.Integer
Dim StandardIndex As System.Integer
Dim FastenerTypeIndex As System.Integer
Dim SSize As System.String
Dim EndType As System.Short
Dim Diameter As System.Double
Dim Depth As System.Double
Dim Length As System.Double
Dim Value1 As System.Double
Dim Value2 As System.Double
Dim Value3 As System.Double
Dim Value4 As System.Double
Dim Value5 As System.Double
Dim Value6 As System.Double
Dim Value7 As System.Double
Dim Value8 As System.Double
Dim Value9 As System.Double
Dim Value10 As System.Double
Dim Value11 As System.Double
Dim Value12 As System.Double
Dim ThreadClass As System.String
Dim RevDir As System.Boolean
Dim FeatureScope As System.Boolean
Dim AutoSelect As System.Boolean
Dim AssemblyFeatureScope As System.Boolean
Dim AutoSelectComponents As System.Boolean
Dim PropagateFeatureToParts As System.Boolean
Dim value As Feature

```

```

value = instance.HoleWizard5(GenericHoleType, StandardIndex, FastenerTypeIndex, SSize, EndType,
Diameter, Depth, Length, Value1, Value2, Value3, Value4, Value5, Value6, Value7, Value8, Value9,
Value10, Value11, Value12, ThreadClass, RevDir, FeatureScope, AutoSelect, AssemblyFeatureScope,
AutoSelectComponents, PropagateFeatureToParts)

```

C#

```

Feature HoleWizard5(
    System.int GenericHoleType,
    System.int StandardIndex,
    System.int FastenerTypeIndex,
    System.string SSize,
    System.short EndType,
    System.double Diameter,
    System.double Depth,
    System.double Length,
    System.double Value1,
    System.double Value2,
    System.double Value3,

```

```

System.double Value4,
System.double Value5,
System.double Value6,
System.double Value7,
System.double Value8,
System.double Value9,
System.double Value10,
System.double Value11,
System.double Value12,
System.string ThreadClass,
System.bool RevDir,
System.bool FeatureScope,
System.bool AutoSelect,
System.bool AssemblyFeatureScope,
System.bool AutoSelectComponents,
System.bool PropagateFeatureToParts
)

```

C++/CLI

```

Feature^ HoleWizard5(
& System.int GenericHoleType,
& System.int StandardIndex,
& System.int FastenerTypeIndex,
& System.String^ SSize,
& System.short EndType,
& System.double Diameter,
& System.double Depth,
& System.double Length,
& System.double Value1,
& System.double Value2,
& System.double Value3,
& System.double Value4,
& System.double Value5,
& System.double Value6,
& System.double Value7,
& System.double Value8,
& System.double Value9,
& System.double Value10,
& System.double Value11,
& System.double Value12,
& System.String^ ThreadClass,
& System.bool RevDir,
& System.bool FeatureScope,
& System.bool AutoSelect,
& System.bool AssemblyFeatureScope,
& System.bool AutoSelectComponents,
& System.bool PropagateFeatureToParts
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

GenericHoleType

Type of hole or slot as defined in [swWzdGeneralHoleTypes_e](#)

StandardIndex

Hole or slot standard property as defined in [swWzdHoleStandards_e](#) (see **Remarks** for legacy holes)

FastenerTypeIndex

Hole or slot fastener type as defined in [swWzdHoleStandardFastenerTypes_e](#) (see **Remarks**)

SSize

Size of the hole or slot (see **Remarks**)

EndType

Hole or slot end type as defined in [swEndConditions_e](#) (see **Remarks**)

Diameter

Diameter of the hole or slot

Depth

Depth of the hole or slot

Length

Length of slot; valid only if *GenericHoleType* set to:

- [swWzdGeneralHoleTypes_e.swWzdCounterBoreSlot](#)
- [swWzdGeneralHoleTypes_e.swWzdCounterSinkSlot](#)
- [swWzdGeneralHoleTypes_e.swWzdHoleSlot](#)

For both straight/pipe and tapered tap holes, the macro recorder fails to properly record this parameter. Before running a macro recording, you must add -1 at this position and delete a -1 before *ThreadClass*.

Value1

Hole or slot parameter

Value2

Hole or slot parameter

Value3

Hole or slot parameter

Value4

Hole or slot parameter
Value5
Hole or slot parameter
Value6
Hole or slot parameter
Value7
Hole or slot parameter
Value8
Hole or slot parameter
Value9
Hole or slot parameter
Value10
Hole or slot parameter
Value11
Hole or slot parameter
Value12
Hole or slot parameter
ThreadClass

One of the following thread classes:

- 1B
- 2B
- 3B

NOTE: Applies to ANSI inch standard only.

RevDir

True to reverse the direction of the hole or slot, false to not

FeatureScope

True if the feature only affects selected bodies, false if the feature affects all bodies

AutoSelect

True to automatically select all bodies and have the feature affect those bodies, false to select the bodies the feature affects

AssemblyFeatureScope

True if the assembly feature only affects selected components in the assembly, false if the assembly feature affects all components in the assembly

AutoSelectComponents

True to automatically select all affected components, false to only use the selected components

PropagateFeatureToParts

True to propagate the assembly feature to the components that the assembly feature affects in the model, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::HoleWizard5](#).

Example

[Create Hole Wizard Hole \(C#\)](#)
[Create Hole Wizard Hole \(VB.NET\)](#)
[Create Hole Wizard Hole \(VBA\)](#)
[Insert Hole Wizard Slot and Hole \(C#\)](#)
[Insert Hole Wizard Slot and Hole \(VB.NET\)](#)
[Insert Hole Wizard Slot and Hole \(VBA\)](#)

Remarks

To create simple holes and slots, read [Hole Wizard Features and WizardHoleFeatureData2 Objects](#). To create Advanced Holes of stacked hole elements, use [IFeatureManager::AdvancedHole](#).

Valid types of slots are counterbore, countersink, or straight.

To add a hole or slot at one or more locations, call [IModelDocExtension::SelectByRay](#) with Mark = 0 for each location.

Before creating a hole or slot using a reference plane and this method, determine the value of the RevDir parameter as follows:

- If the material is on one side of the reference plane, then RevDir is false.
- If the material is on both sides of the reference plane, then compare the distances between the reference plane and the material on both sides. If the distance between the reference plane and the material along its normal direction is bigger, then RevDir is true; otherwise, RevDir is false.

You can also use [IModelDoc2::RayIntersections](#) and [IModelDoc2::GetRayIntersectionsPoints](#) to find out if the material is on one side or both sides and to compare the distances.

Parameters for this method:

- The FastenerTypeIndex parameter has to match the standard and be valid for that hole or slot type, or an error occurs.
- The SSize parameter must be a valid size for the fastener type, or an error occurs.
- If EndType is 10 = swEndCondUpToSelection (supersedes swEndCondUpToSurface and swEndCondUpToVertex), then you must use [IModelDocExtension::SelectById2](#) with Mark = 2 to pre-select the surface or vertex. The macro recorder fails to properly record EndType for Up To Surface and Up To Vertex end conditions. Instead of swEndCondUpToSelection=10, it records 4 and 3. Before running an Up to Surface or Vertex macro recording, you must change EndType to 10. The macro recorder also fails to properly record the Up To Next end condition. Instead of swEndCondUpToNext=11, it records swEndCondThroughNext=2. It is advisable to record only Blind or Through All end conditions.
- Screw fit is defined as [swWzdHoleScrewClearanceTypes_e](#) and is only used if a hole or slot diameter is not specified; default value is NORMAL_FIT.
- Drill angle defaults to 118 degrees (2.05948851735331 radians), if not specified.
- Input angles are in radians; input values are in meters.
- The Value1 through Value12 parameters are different for each type of hole or slot. The possible values are as follows, organized by hole and slot type. SOLIDWORKS ignores parameters set to -1.

Counterbore Holes and Slots

Head Clearance - Added to either the input counterbore depth or the counterbore depth in the standard.

Screw Fit - Value from [swWzdHoleScrewClearanceTypes_e](#)

Parameters for all counterbore holes and counterbore slots:

1. CounterBore Diameter
2. CounterBore Depth
3. Head Clearance
4. Screw Fit
5. Drill Angle at Bottom (valid only if EndType is *not* swEndCondUpToNext or swEndCondThroughAll)
6. Near Csink Diameter
7. Near Csink Angle
8. Underhead Csink Diameter
9. Underhead Csink Angle
10. Far Csink Diameter
11. Far Csink Angle
12. Offset (valid only if EndType is swEndCondOffsetFromSurface)

Countersink Holes and Slots

Head Clearance Type - Value from [swWzdHoleCounterSinkHeadClearanceTypes_e](#), which represents how to add the head clearance values to the hole or slot.

Screw Fit - Value from [swWzdHoleScrewClearanceTypes_e](#)

Parameters for all countersink holes and countersink slots:

1. Near Csink Diameter
2. Near Csink Angle
3. Head Clearance
4. Screw Fit
5. Drill Angle at Bottom (valid only if EndType is *not* swEndCondUpToNext or swEndCondThroughAll)
6. Far Csink Diameter
7. Far Csink Angle
8. Offset (valid only if EndType is swEndCondOffsetFromSurface)
9. Head Clearance Type
10. -1
11. -1
12. -1

Regular Holes and Straight Slots

Screw Fit - Value from [swWzdHoleScrewClearanceTypes_e](#)

Parameters for all regular holes and straight slots:

1. Screw Fit
2. Drill Angle at Bottom (valid only if EndType is *not* swEndCondUpToNext or swEndCondThroughAll)
3. Near Csink Diameter
4. Near Csink Angle
5. Far Csink Diameter
6. Far Csink Angle
7. Offset (valid only if EndType is swEndCondOffsetFromSurface)
8. -1
9. -1
10. -1
11. -1
12. -1

Pipe/Straight Tap Holes

Tap Thread Depth - If the user specifies thread depth, SOLIDWORKS always uses this depth. Tap thread depths can be calculated:

1. Helicoil holes: calculated based on a constant (determined by hole type; i.e., insert = 1.0 * diameter) + thread depth from the standard * thread major diameter.
2. Tapped holes: calculated based on 2 * thread diameter.

Cosmetic Thread Type - Value from [swWzdHoleCosmeticThreadTypes_e](#); defaults to swCosmeticThreadNone

Near Countersink Angle - Can be calculated from standard data as a default

Hcoil Tap Type - Value from [swWzdHoleHcoilTapTypes_e](#); defaults to swTapTypePlug

Thread End Condition - Value from [swWzdHoleThreadEndCondition_e](#); defaults to swEndThreadTypeBLIND

Parameters for all pipe/straight tap holes:

1. Tap Thread Depth
2. Near Csink Diameter
3. Near Csink Angle
4. Far Csink Diameter
5. Far Csink Angle
6. Drill Angle at Bottom (valid only if EndType is *not* swEndCondUpToNext or swEndCondThroughAll)
7. Cosmetic Thread Type
8. Thread End Condition (not valid if StandardIndex is swStandardHelicoilInch or swStandardHelicoilMetric)
9. Helicoil Tap Type (valid only if StandardIndex is swStandardHelicoilInch or swStandardHelicoilMetric)
10. Offset (valid only if EndType is swEndCondOffsetFromSurface)
11. -1
12. -1

Tapered Tap Holes

NOTE: The macro recorder fails to properly record tapered tap hole parameters. Follow the instructions in this help to correct recordings before playing them.

Cosmetic Thread Type - Specified as defined in [swWzdHoleCosmeticThreadTypes_e](#); defaults to swCosmeticThreadNone

Parameters for all tapered tap holes:

1. Tap Thread Depth
2. Near Csink Diameter
3. Near Csink Angle
4. Far Csink Diameter

5. Far Csink Angle
6. Drill Angle at Bottom (valid only if EndType is *not* swEndCondUpToNext or swEndCondThroughAll)
7. Cosmetic Thread Type
8. Offset (valid only if EndType is swEndCondOffsetFromSurface)
9. -1
10. -1
11. -1
12. -1

Legacy Holes

StandardIndex, FastenerTypeIndex, and SSize are:

- Not relevant
- SOLIDWORKS ignores them

Value1 - Value12 for different types of legacy holes. Use -1 for unused values:

Simple

1. Diameter
2. Depth
3. Sketch ID

Tapered

1. Minor Diameter
2. Depth
3. Sketch ID
4. Major Diameter

Counterbored

1. Diameter
2. Depth
3. Sketch ID
4. C-Bore Diameter
5. C-Bore Depth

Countersunk

1. Diameter
2. Depth
3. Sketch ID
4. C-Sink Angle
5. C-Sink Diameter

Counterdrilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Drill Diameter
5. C-Drill Depth
6. C-Drill Angle

Simple Drilled

1. Diameter
2. Depth
3. Sketch ID
4. Drill Angle

Tapered Drilled

1. Minor Diameter
2. Depth
3. Sketch ID
4. Major Diameter
5. Drill Angle

C-Bored Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Bore Diameter
5. C-Bore Depth
6. Drill Angle

C-Sunk Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Sink Diameter
5. C-Sink Angle
6. Drill Angle

C-Drilled Drilled

1. Diameter
2. Depth
3. Sketch ID
4. C-Drill Diameter
5. C-Drill Depth
6. Drill Angle
7. C-Drill Angle

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IWizardHoleFeatureData2 Interface](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : IFeatureFillet Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.IFeatureFillet2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureFillet( _  
    ByVal Options As System.Integer, _  
    ByVal R1 As System.Double, _  
    ByVal Ftyp As System.Integer, _  
    ByVal OverflowType As System.Integer, _  
    ByVal NRadii As System.Integer, _  
    ByRef Radii As System.Double, _  
    ByVal SetbackDistCount As System.Integer, _  
    ByRef SetBackDistances As System.Double, _  
    ByVal PointCount As System.Integer, _  
    ByRef PointRadiusArray As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Options As System.Integer  
Dim R1 As System.Double  
Dim Ftyp As System.Integer  
Dim OverflowType As System.Integer  
Dim NRadii As System.Integer  
Dim Radii As System.Double  
Dim SetbackDistCount As System.Integer  
Dim SetBackDistances As System.Double  
Dim PointCount As System.Integer  
Dim PointRadiusArray As System.Double  
Dim value As Feature
```

```
value = instance.IFeatureFillet(Options, R1, Ftyp, OverflowType, NRadii, Radii, SetbackDistCount,  
SetBackDistances, PointCount, PointRadiusArray)
```

C#

```
Feature IFeatureFillet(  
    System.int Options,  
    System.double R1,  
    System.int Ftyp,  
    System.int OverflowType,  
    System.int NRadii,  
    ref System.double Radii,  
    System.int SetbackDistCount,  
    ref System.double SetBackDistances,  
    System.int PointCount,  
    ref System.double PointRadiusArray  
)
```

C++/CLI

```
Feature^ IFeatureFillet(  
& System.int Options,  
& System.double R1,  
& System.int Ftyp,  
& System.int OverflowType,  
& System.int NRadii,  
& System.double% Radii,  
& System.int SetbackDistCount,  
& System.double% SetBackDistances,  
& System.int PointCount,  
& System.double% PointRadiusArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options
Feature fillet options as defined in [swFeatureFilletOptions_e](#)
R1
Radius for uniform radius edge fillet
Ftyp
Type of fillet as defined in [swFeatureFilletType_e](#)
OverflowType
Control of fillet overflowing onto adjacent surfaces as defined in [swFilletOverFlowType_e](#)
NRadii
Number of radii for variable radius fillet
Radii

- in-process, unmanaged C++: Pointer to an array of doubles containing the radii for the variable radius fillet
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

SetbackDistCount
Number of setback distances for the fillet along the edge
SetBackDistances

- in-process, unmanaged C++: Pointer to an array of doubles containing setback distances for the fillet along the edge of size SetbackDistCount
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

PointCount
Number of control point radii at various points along the length of the edge
PointRadiusArray

- in-process, unmanaged C++: Pointer to an array of doubles containing the control point radii of size PointCount
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

[IFeature](#) pointer

Remarks

See the [IFeatureManager::FeatureFillet](#) Remarks.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::FilletXpertChange Method](#)
[IFeatureManager::FilletXpertRemove Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IFeatureFillet2 Method (IFeatureManager)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IFeatureManager Interface](#) : IFeatureFillet2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::FeatureFillet3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IFeatureFillet2( _  
    ByVal Options As System.Integer, _  
    ByVal R1 As System.Double, _  
    ByVal Rho As System.Double, _  
    ByVal Ftyp As System.Integer, _  
    ByVal OverflowType As System.Integer, _  
    ByVal ConicRhoType As System.Integer, _  
    ByVal NRadii As System.Integer, _  
    ByRef Radii As System.Double, _  
    ByRef RhoArr As System.Double, _  
    ByVal SetbackDistCount As System.Integer, _  
    ByRef SetBackDistances As System.Double, _  
    ByVal PointCount As System.Integer, _  
    ByRef PointRadiusArray As System.Double, _  
    ByRef PointRhoArray As System.Double _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim Options As System.Integer  
Dim R1 As System.Double
```

```

Dim Rho As System.Double
Dim Ftyp As System.Integer
Dim OverflowType As System.Integer
Dim ConicRhoType As System.Integer
Dim NRadii As System.Integer
Dim Radii As System.Double
Dim RhoArr As System.Double
Dim SetbackDistCount As System.Integer
Dim SetBackDistances As System.Double
Dim PointCount As System.Integer
Dim PointRadiusArray As System.Double
Dim PointRhoArray As System.Double
Dim value As Feature

```

```

value = instance.IFeatureFillet2(Options, R1, Rho, Ftyp, OverflowType, ConicRhoType, NRadii, Radii, RhoArr, SetbackDistCount, SetBackDistances, PointCount, PointRadiusArray, PointRhoArray)

```

C#

```

Feature IFeatureFillet2(
    System.int Options,
    System.double R1,
    System.double Rho,
    System.int Ftyp,
    System.int OverflowType,
    System.int ConicRhoType,
    System.int NRadii,
    ref System.double Radii,
    ref System.double RhoArr,
    System.int SetbackDistCount,
    ref System.double SetBackDistances,
    System.int PointCount,
    ref System.double PointRadiusArray,
    ref System.double PointRhoArray
)

```

C++/CLI

```

Feature^ IFeatureFillet2(
& System.int Options,
& System.double R1,
& System.double Rho,
& System.int Ftyp,
& System.int OverflowType,
& System.int ConicRhoType,
& System.int NRadii,
& System.double% Radii,
& System.double% RhoArr,
& System.int SetbackDistCount,
& System.double% SetBackDistances,
& System.int PointCount,
& System.double% PointRadiusArray,
& System.double% PointRhoArray
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options

Feature fillet options as defined in [swFeatureFilletOptions_e](#)

R1

Uniform radius of the fillet; valid only if:

- Ftyp != [swFeatureFilletType_e.swFeatureFilletType_VariableRadius](#)
- Options include [swFeatureFilletOptions_e.swFeatureFilletUniformRadius](#)

Rho

Value that determines the conic shape of the fillet:

- Conic rho value, if ConicRhoType = [swFeatureFilletProfileType_e.swFeatureFilletConicRho](#)
- Conic radius value, if ConicRhoType = [swFeatureFilletProfileType_e.swFeatureFilletConicRadius](#)
- Circular radius value, if ConicRhoType = [swFeatureFilletProfileType_e.swFeatureFilletCircular](#)

Ftyp

Type of fillet as defined in [swFeatureFilletType_e](#) (see **Remarks**)

OverflowType

Control of fillet overflowing onto adjacent surfaces as defined in [swFilletOverFlowType_e](#)

ConicRhoType

Conic fillet profile type as defined in [swFeatureFilletProfileType_e](#)

NRadii

Number of elements in the Radii array

Radii

- In-process, unmanaged C++: Pointer to an array containing the radii for the variable radius fillet; valid only if:

- Ftyp = [swFeatureFilletType_e](#).swFeatureFilletType_VariableRadius
- Options include [swFeatureFilletOptions_e](#).swFeatureFilletVarRadiusType
- Options do not include [swFeatureFilletOptions_e](#).swFeatureFilletUniformRadius
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

RhoArr

- In-process, unmanaged C++: Pointer to an array of conic shape values for the specified ConicRhoType for the variable radius fillet; valid only if:
 - Ftyp = [swFeatureFilletType_e](#).swFeatureFilletType_VariableRadius
 - Options include [swFeatureFilletOptions_e](#).swFeatureFilletVarRadiusType
 - Options do not include [swFeatureFilletOptions_e](#).swFeatureFilletUniformRadius
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

SetbackDistCount

Number of elements in the SetBackDistances array

SetBackDistances

- In-process, unmanaged C++: Pointer to an array containing setback distances along the fillet edge
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

PointCount

Number of elements in the PointRadiusArray and PointRhoArray arrays

PointRadiusArray

- In-process, unmanaged C++: Pointer to an array containing radii at various control points along the length of the edge; valid only if Ftyp = [swFeatureFilletType_e](#).swFeatureFilletType_VariableRadius
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

PointRhoArray

- In-process, unmanaged C++: Pointer to an array containing conic shape values for the specified ConicRhoType at various control points along the length of the edge; valid only if Ftyp = [swFeatureFilletType_e](#).swFeatureFilletType_VariableRadius
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

[IFeature](#) pointer

Remarks

See the [IFeatureManager::FeatureFillet2](#) Remarks.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::FeatureFillet2 Method \(\)](#)
[ISimpleFilletFeatureData2 Interface](#)
[IVariableFilletFeatureData2 Interface](#)
[IFeatureManager::FilletXpertChange Method \(\)](#)
[IFeatureManager::FilletXpertMakeCorner Method \(\)](#)
[IFeatureManager::FilletXpertRemove Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFeatures Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : IGetFeatures Method (IFeatureManager)

Gets the features in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFeatures( _
    ByVal ToplevelOnly As System.Boolean, _
    ByVal Count As System.Integer _
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim ToplevelOnly As System.Boolean
Dim Count As System.Integer
Dim value As Feature
```

```
value = instance.IGetFeatures(ToplevelOnly, Count)
```

C#

```
Feature IGetFeatures(  
    System.bool TopLevelOnly,  
    System.int Count  
)
```

C++/CLI

```
Feature^ IGetFeatures(  
& System.bool TopLevelOnly,  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopLevelOnly

True to get only the features at the top level of the FeatureManager design tree, false to get the top-level features and all child features in the FeatureManager design tree

Count

Number of features

Return Value

- in-process, unmanaged C++: Pointer to an array of [IFeature](#) objects
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [IFeatureManager::GetFeatureCount](#) before calling this method to get the size for the array of features.

If TopLevelOnly is false, then this method gets all of the feature and child features in this document. It does not get features in components.

The features that are returned by this method can be in any order. You should not rely on the order to indicate anything about children or parents. If hierarchy and order information is needed, then use [IPartDoc::FirstFeature](#) or [IPartDoc::IFirstFeature](#) or [IModelDoc2::FirstFeature](#) or [IModelDoc2::FirstFeature](#), [IFeature::GetNextFeature](#) or [IModelDoc2::GetNextFeature](#), [IFeature::GetFirstSubFeature](#) or [IFeature::IGetFirstSubFeature](#), and [IFeature::GetNextSubFeature](#) or [IFeature::IGetNextSubFeature](#) to retrieve your information.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::GetFeatures Method](#)
[IFeature::GetChildren Method](#)
[IFeature::GetOwnerFeature Method](#)
[IFeature::GetParents Method](#)
[IFeature::IGetChildCount Method](#)
[IFeature::IGetChildren Method](#)
[IFeature::IGetParentCount Method](#)
[IFeature::IGetParents Method](#)
[IComponent2::FirstFeature Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCombineFeature Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCombineFeature Method (IFeatureManager)

Combines the specified bodies in the multibody part to create a combine feature.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertCombineFeature( _  
    ByVal OperationType As System.Integer, _  
    ByVal MainBody As Body2, _  
    ByVal ToolsCount As System.Integer, _  
    ByRef ToolsArr As Body2 _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim OperationType As System.Integer
Dim MainBody As Body2
Dim ToolsCount As System.Integer
Dim ToolsArr As Body2
Dim value As Feature

value = instance.IInsertCombineFeature(OperationType, MainBody, ToolsCount, ToolsArr)
```

C#

```
Feature IInsertCombineFeature(
    System.int OperationType,
    Body2 MainBody,
    System.int ToolsCount,
    ref Body2 ToolsArr
)
```

C++/CLI

```
Feature^ IInsertCombineFeature(
& System.int OperationType,
& Body2^ MainBody,
& System.int ToolsCount,
& Body2^% ToolsArr
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType
Operation as defined in [swBodyOperationType_e](#)

MainBody

If <i>OperationType</i> is ...	Then set <i>MainBody</i> to ...
swBodyOperationType_e_SWBODYADD	Nothing or null
swBodyOperationType_e_SWBODYCUT	Target body
swBodyOperationType_e_SWBODYINTERSECT	Nothing or null

(See **Remarks**)
ToolsCount
Number of bodies to combine
ToolsArr

- in-process, unmanaged C++: Pointer to an array of [bodies](#) to combine of size *ToolsCount* (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

[IFeature](#)

 Remarks

You can either call this method, directly passing in the bodies with *MainBody* and *ToolVar*, or you can:

1. Select the bodies using [IModelDocExtension::SelectByID2](#) as follows:

To select...	Use Mark...
MainBody	1
ToolVar bodies	2

2. Call this method, setting *MainBody* to Nothing or null and *ToolVar* to an empty array.

See the SOLIDWORKS Help for more information about the combined feature.

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertCombineFeature Method](#)
[ICombineBodiesFeatureData Interface](#)

 Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IFeatureManager Interface : InsertMacroFeature Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMacroFeature3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertMacroFeature( _  
    ByVal BaseName As System.String, _  
    ByVal ProgId As System.String, _  
    ByRef MacroMethods As System.String, _  
    ByVal ParamCount As System.Integer, _  
    ByRef ParamNames As System.String, _  
    ByRef ParamTypes As System.Integer, _  
    ByRef ParamValues As System.String, _  
    ByVal EditBody As Body2, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BaseName As System.String  
Dim ProgId As System.String  
Dim MacroMethods As System.String  
Dim ParamCount As System.Integer  
Dim ParamNames As System.String  
Dim ParamTypes As System.Integer  
Dim ParamValues As System.String  
Dim EditBody As Body2  
Dim Options As System.Integer  
Dim value As Feature  
  
value = instance.IInsertMacroFeature(BaseName, ProgId, MacroMethods, ParamCount, ParamNames,  
ParamTypes, ParamValues, EditBody, Options)
```

C#

```
Feature IInsertMacroFeature(  
    System.string BaseName,  
    System.string ProgId,  
    ref System.string MacroMethods,  
    System.int ParamCount,  
    ref System.string ParamNames,  
    ref System.int ParamTypes,  
    ref System.string ParamValues,  
    Body2 EditBody,  
    System.int Options  
)
```

C++/CLI

```
Feature^ IInsertMacroFeature(  
& System.String^ BaseName,  
& System.String^ ProgId,  
& System.String^% MacroMethods,  
& System.int ParamCount,  
& System.String^% ParamNames,  
& System.int% ParamTypes,  
& System.String^% ParamValues,  
& Body2^ EditBody,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BaseName

ProgId

MacroMethods

ParamCount
ParamNames
ParamTypes
ParamValues
EditBody
Options

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertMacroFeature](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMacroFeature2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMacroFeature2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertMacroFeature3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertMacroFeature2( _  
    ByVal BaseName As System.String, _  
    ByVal ProgId As System.String, _  
    ByRef MacroMethods As System.String, _  
    ByVal ParamCount As System.Integer, _  
    ByRef ParamNames As System.String, _  
    ByRef ParamTypes As System.Integer, _  
    ByRef ParamValues As System.String, _  
    ByVal DimCount As System.Integer, _  
    ByRef DimTypes As System.Integer, _  
    ByRef DimCountValues As System.Double, _  
    ByVal EditBody As Body2, _  
    ByVal IconCount As System.Integer, _  
    ByRef IconFiles As System.String, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim BaseName As System.String  
Dim ProgId As System.String  
Dim MacroMethods As System.String  
Dim ParamCount As System.Integer  
Dim ParamNames As System.String  
Dim ParamTypes As System.Integer  
Dim ParamValues As System.String  
Dim DimCount As System.Integer  
Dim DimTypes As System.Integer  
Dim DimCountValues As System.Double  
Dim EditBody As Body2  
Dim IconCount As System.Integer  
Dim IconFiles As System.String  
Dim Options As System.Integer  
Dim value As Feature  
  
value = instance.IInsertMacroFeature2(BaseName, ProgId, MacroMethods, ParamCount, ParamNames,  
ParamTypes, ParamValues, DimCount, DimTypes, DimCountValues, EditBody, IconCount, IconFiles, Options)
```

C#

```

Feature IInsertMacroFeature2(
    System.string BaseName,
    System.string ProgId,
    ref System.string MacroMethods,
    System.int ParamCount,
    ref System.string ParamNames,
    ref System.int ParamTypes,
    ref System.string ParamValues,
    System.int DimCount,
    ref System.int DimTypes,
    ref System.double DimCountValues,
    Body2 EditBody,
    System.int IconCount,
    ref System.string IconFiles,
    System.int Options
)

```

C++/CLI

```

Feature^ IInsertMacroFeature2(
& System.String^ BaseName,
& System.String^ ProgId,
& System.String^% MacroMethods,
& System.int ParamCount,
& System.String^% ParamNames,
& System.int% ParamTypes,
& System.String^% ParamValues,
& System.int DimCount,
& System.int% DimTypes,
& System.double% DimCountValues,
& Body2^ EditBody,
& System.int IconCount,
& System.String^% IconFiles,
& System.int Options
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BaseName

ProgId

MacroMethods

ParamCount

ParamNames

ParamTypes

ParamValues

DimCount

DimTypes

DimCountValues

EditBody

IconCount

IconFiles

Options

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::IInsertMacroFeature2](#).

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)



Inserts a macro feature in this model.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertMacroFeature3( _  
    ByVal BaseName As System.String, _  
    ByVal ProgId As System.String, _  
    ByRef MacroMethods As System.String, _  
    ByVal ParamCount As System.Integer, _  
    ByRef ParamNames As System.String, _  
    ByRef ParamTypes As System.Integer, _  
    ByRef ParamValues As System.String, _  
    ByVal DimCount As System.Integer, _  
    ByRef DimTypes As System.Integer, _  
    ByRef DimCountValues As System.Double, _  
    ByVal BodyCount As System.Integer, _  
    ByRef EditBodies As Body2, _  
    ByVal IconCount As System.Integer, _  
    ByRef IconFiles As System.String, _  
    ByVal Options As System.Integer _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim BaseName As System.String  
Dim ProgId As System.String  
Dim MacroMethods As System.String  
Dim ParamCount As System.Integer  
Dim ParamNames As System.String  
Dim ParamTypes As System.Integer  
Dim ParamValues As System.String  
Dim DimCount As System.Integer  
Dim DimTypes As System.Integer  
Dim DimCountValues As System.Double  
Dim BodyCount As System.Integer  
Dim EditBodies As Body2  
Dim IconCount As System.Integer  
Dim IconFiles As System.String  
Dim Options As System.Integer  
Dim value As Feature  
  
value = instance.IInsertMacroFeature3(BaseName, ProgId, MacroMethods, ParamCount, ParamNames,  
ParamTypes, ParamValues, DimCount, DimTypes, DimCountValues, BodyCount, EditBodies, IconCount,  
IconFiles, Options)
```

C#

```
Feature IInsertMacroFeature3(  
    System.string BaseName,  
    System.string ProgId,  
    ref System.string MacroMethods,  
    System.int ParamCount,  
    ref System.string ParamNames,  
    ref System.int ParamTypes,  
    ref System.string ParamValues,  
    System.int DimCount,  
    ref System.int DimTypes,  
    ref System.double DimCountValues,  
    System.int BodyCount,  
    ref Body2 EditBodies,  
    System.int IconCount,  
    ref System.string IconFiles,  
    System.int Options  
)
```

C++/CLI

```
Feature^ IInsertMacroFeature3(  
& System.String^ BaseName,  
& System.String^ ProgId,  
& System.String^% MacroMethods,  
& System.int ParamCount,  
& System.String^% ParamNames,  
& System.int% ParamTypes,  
& System.String^% ParamValues,  
& System.int DimCount,  
& System.int% DimTypes,  
& System.double% DimCountValues,  
& System.int BodyCount,  
& Body2^% EditBodies,  
& System.int IconCount,  
& System.String^% IconFiles,  
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

- BaseName*
Name of the base feature (see **Remarks**)
- ProgId*
Indicates callback object (see **Remarks**)
- MacroMethods*
Null; valid only for VBA
- ParamCount*
Number of parameters
- ParamNames*
Array of strings of the parameters of size ParamCount
- ParamTypes*
Array of the types of parameters of size paramCount as defined by [swMacroFeatureParamType_e](#)
- ParamValues*
Array of strings of the values of parameters of size ParamCount
- DimCount*
Number of dimensions
- DimTypes*
Array of the types of dimensions as defined by [swDimensionType_e](#) (see **Remarks**)
- DimCountValues*
Array of doubles of the values of dimensions of size DimCount
- BodyCount*
Number of bodies to modify in the macro feature
- EditBodies*
Array of [IBody2](#) objects to modify in the macro feature
- IconCount*
Number of icons
- IconFiles*
Array of strings of the file names for the icons (see **Remarks**)
- Options*
Placement of the macro feature in the FeatureManager design tree as defined by [swMacroFeatureOptions_e](#) (see **Remarks**)

Return Value

Pointer to the [IFeature](#) object

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertMacroFeature3](#).

 Remarks

BaseName

- The argument BaseName is serialized within the feature and cannot be changed.
- You can find out the name of the base feature by using [IMacroFeatureData::GetBaseName](#).
- The BaseName argument is also used to generate the name of the feature when the feature is first created.

ProgId

If creating a macro feature using...	Then ProgId is...
COM callback methods	The name of the component that implements the COM callback methods. The object that is repi by ProgID must support the ISwComFeature interface. InsertMacroFeature3 ("Sample", "Sample.MyFeature", _ Nothing, paramNames, paramTypes, _ paramValues, dimTypes, dimValues, _ editBodies, iconFiles, _ swMacroFeatureByDefault) In the COM server, the Sample.Feature class is derived or supports the ISwComFeature interfa These methods support the macro feature's rebuild, edit, and security functions.

DimTypes

Only these dimension types are supported by this method:

- swAngularDimension
- swLinearDimension
- swRadialDimension

IconFiles

The array of the file names for the icons can contain either three or nine strings.

Number of strings in array	Types of images in this order	Image format and sizes
Three	1. Regular 2. Suppressed 3. Highlighted	• Windows bitmap (*.bmp) format • 16 pixels wide X 18 pixels high
Nine	1. Regular small 2. Suppressed small 3. Highlighted small 4. Regular medium 5. Suppressed medium 6. Highlighted medium 7. Regular large 8. Suppressed large 9. Highlighted large	• Windows bitmap (*.bmp) format • Recommended sizes are: • Small: 20 pixels wide X 20 pixels high • Medium: 32 pixels wide X 32 pixels high • Large: 40 pixels wide X 40 pixels high
NOTES: • This size array is only valid for macro features created in parts, assemblies, and drawings in SOLIDWORKS 2017 and later. • SOLIDWORKS displays the appropriate images based on the current DPI setting of the display device.		

You can specify either the full path name or just the file name for the strings; for example, c:\bitmaps\icon1.bmp or icon1.bmp.

Options

swMacroFeatureOptions_e.swMacroFeatureEmbedMacroFile is not supported by programming languages for the Microsoft .NET Framework; for example, not supported by C#, Visual Basic .NET, or Managed C++.

See [Overview of Macro Features](#) for more information.

See Also

- [IFeatureManager Interface](#)
- [IFeatureManager Members](#)
- [IMacroFeatureData Interface](#)
- [IFeatureManager.InsertMacroFeature3 Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertMoveFace2 Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertMoveFace2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertMoveFace3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertMoveFace2( _  
    ByVal MoveType As System.Integer, _  
    ByVal ReverseDir As System.Boolean, _  
    ByVal Angle As System.Double, _  
    ByVal Distance As System.Double, _  
    ByRef TranslationParams As System.Double, _  
    ByRef RotationParams As System.Double _  
    ) As Feature
```

Visual Basic (Usage)



Copy Code

```

Dim instance As IFeatureManager
Dim MoveType As System.Integer
Dim ReverseDir As System.Boolean
Dim Angle As System.Double
Dim Distance As System.Double
Dim TranslationParams As System.Double
Dim RotationParams As System.Double
Dim value As Feature

```

```

value = instance.IInsertMoveFace2(MoveType, ReverseDir, Angle, Distance, TranslationParams,
RotationParams)

```

C#

```

Feature IInsertMoveFace2(
    System.int MoveType,
    System.bool ReverseDir,
    System.double Angle,
    System.double Distance,
    ref System.double TranslationParams,
    ref System.double RotationParams
)

```

C++/CLI

```

Feature^ IInsertMoveFace2(
    & System.int MoveType,
    & System.bool ReverseDir,
    & System.double Angle,
    & System.double Distance,
    & System.double% TranslationParams,
    & System.double% RotationParams
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

MoveType

Type of move:

- 0 = Offset
- 1 = Translate
- 2 = Rotate

ReverseDir

True to reverse the direction, false to not

Angle

If MoveType is Rotate, then specify the angle at which to draft the faces

Distance

Distance to offset or translate the faces

TranslationParams

- in-process, unmanaged C++: Pointer to an array three doubles for delta x, delta y, and delta z direction translation
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

RotationParams

- in-process, unmanaged C++: Pointer to an array six doubles:
 - first three doubles are the x, y, and z rotation origin
 - last three doubles are the x, y, and z rotation angle
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Return Value

Pointer to the [IFeature](#) object

Remarks

Use the following marks with [IModelDocExtension::SelectByID2](#):

- 1 = face
- 2 = direction reference (plane, planar face, linear edge, or reference axis) for translate
- 4 = axis reference (linear edge or reference axis) for rotate

If you specify a value for TranslationParams or RotationParams, then do not specify a value for Distance or Angle, respectively. The translation or rotation parameters are calculated internally when Distance or Angle is specified.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IMoveFaceFeatureData Interface](#)
[IFeatureManager.InsertMoveFace2 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertReferencePoint Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertReferencePoint Method (IFeatureManager)

Creates the geometry for the reference points based on any of these selected entities: edges, faces, planes, vertices, or sketch geometry.

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertReferencePoint( _  
    ByVal NRefPointType As System.Integer, _  
    ByVal NRefPointAlongCurveType As System.Integer, _  
    ByVal DDistance\_or\_Percent As System.Double, _  
    ByVal NumberOfRefPoints As System.Integer _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim NRefPointType As System.Integer  
Dim NRefPointAlongCurveType As System.Integer  
Dim DDistance_or_Percent As System.Double  
Dim NumberOfRefPoints As System.Integer  
Dim value As Feature  
  
value = instance.IInsertReferencePoint(NRefPointType, NRefPointAlongCurveType, DDistance_or_Percent,  
NumberOfRefPoints)
```

C#

```
Feature IInsertReferencePoint(  
    System.int NRefPointType,  
    System.int NRefPointAlongCurveType,  
    System.double DDistance\_or\_Percent,  
    System.int NumberOfRefPoints  
)
```

C++/CLI

```
Feature^ IInsertReferencePoint(  
& System.int NRefPointType,  
& System.int NRefPointAlongCurveType,  
& System.double DDistance\_or\_Percent,  
& System.int NumberOfRefPoints  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NRefPointType

Type of reference point as defined by [swRefPointType_e](#)

NRefPointAlongCurveType

Distance, percentage, or evenly distributed as defined by [swRefPointAlongCurveType_e](#)

DDistance_or_Percent

Distance at which to create the reference point on the selected entities or percentage of the length of the selected entities at which to create the reference point if *NRefPointAlongCurveType* is *swRefPointAlongCurveDistance* or *swRefPointAlongCurvePercentage*, respectively

NumberOfRefPoints

Number of reference points to create and evenly distribute on the selected entities if *swRefPointAlongCurveType* is *swRefPointAlongCurveEvenlyDistributed*

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertReferencePoint](#).

Remarks

This method creates one or more reference point features and adds them to the FeatureManager design tree. If the reference point feature is not created, a NULL value is returned.

The NumberOfRefPoints argument must contain a value of 1 to successfully create one reference point feature. Use the NumberOfRefPoints value to allocate memory for the return value array.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertReferencePoint Method](#)
[RefPointFeatureData Interface](#)
[IFeatureManager.EditReferencePoint Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSheetMetalEdgeFlange2 Method (IFeatureManager)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IFeatureManager Interface](#) : InsertSheetMetalEdgeFlange2 Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.CreateDefinition](#) and [IFeatureManager.CreateFeature](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertSheetMetalEdgeFlange2( _  
    ByVal EdgeCount As System.Integer, _  
    ByRef FlangeEdges As Edge, _  
    ByVal SketchFeatCount As System.Integer, _  
    ByRef SketchFeat As Feature, _  
    ByVal BooleanOptions As System.Integer, _  
    ByVal FlangeAngle As System.Double, _  
    ByVal FlangeRadius As System.Double, _  
    ByVal BendPosition As System.Integer, _  
    ByVal FlangeOffsetDist As System.Double, _  
    ByVal ReliefType As System.Integer, _  
    ByVal FlangeReliefRatio As System.Double, _  
    ByVal FlangeReliefWidth As System.Double, _  
    ByVal FlangeReliefDepth As System.Double, _  
    ByVal FlangeSharpType As System.Integer, _  
    ByVal CustomBendAllowance As CustomBendAllowance _  
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim EdgeCount As System.Integer  
Dim FlangeEdges As Edge  
Dim SketchFeatCount As System.Integer  
Dim SketchFeat As Feature  
Dim BooleanOptions As System.Integer  
Dim FlangeAngle As System.Double  
Dim FlangeRadius As System.Double  
Dim BendPosition As System.Integer  
Dim FlangeOffsetDist As System.Double  
Dim ReliefType As System.Integer  
Dim FlangeReliefRatio As System.Double  
Dim FlangeReliefWidth As System.Double  
Dim FlangeReliefDepth As System.Double  
Dim FlangeSharpType As System.Integer  
Dim CustomBendAllowance As CustomBendAllowance  
Dim value As Feature
```




```
value = instance.IInsertSheetMetalEdgeFlange2(EdgeCount, FlangeEdges, SketchFeatCount, SketchFeat, BooleanOptions, FlangeAngle, FlangeRadius, BendPosition, FlangeOffsetDist, ReliefType, FlangeReliefRatio, FlangeReliefWidth, FlangeReliefDepth, FlangeSharpType, CustomBendAllowance)
```

C#

```
Feature IInsertSheetMetalEdgeFlange2(  
    System.int EdgeCount,  
    ref Edge FlangeEdges,  
    System.int SketchFeatCount,  
    ref Feature SketchFeat,  
    System.int BooleanOptions,  
    System.double FlangeAngle,  
    System.double FlangeRadius,  
    System.int BendPosition,  
    System.double FlangeOffsetDist,  
    System.int ReliefType,  
    System.double FlangeReliefRatio,  
    System.double FlangeReliefWidth,  
    System.double FlangeReliefDepth,  
    System.int FlangeSharpType,  
    CustomBendAllowance CustomBendAllowance  
)
```

C++/CLI

```
Feature^ IInsertSheetMetalEdgeFlange2(  
& System.int EdgeCount,  
& Edge^% FlangeEdges,  
& System.int SketchFeatCount,  
& Feature^% SketchFeat,  
& System.int BooleanOptions,  
& System.double FlangeAngle,  
& System.double FlangeRadius,  
& System.int BendPosition,  
& System.double FlangeOffsetDist,  
& System.int ReliefType,  
& System.double FlangeReliefRatio,  
& System.double FlangeReliefWidth,  
& System.double FlangeReliefDepth,  
& System.int FlangeSharpType,  
& CustomBendAllowance^ CustomBendAllowance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EdgeCount

Number of edges for the flange

FlangeEdges

Array of edge to which to apply a flange

SketchFeatCount

Number of sketches for the flange

SketchFeat

Array of sketch [features](#) to use for the flange

BooleanOptions

Flange options as defined by [swInsertEdgeFlangeOptions_e](#)

FlangeAngle

Flange angle

FlangeRadius

Bend radius

BendPosition

Flange bend position as defined by [swFlangePositionTypes_e](#)

FlangeOffsetDist

Length of flange

ReliefType

Relief type as defined by [swSheetMetalReliefTypes_e](#)

FlangeReliefRatio

Relief ratio

FlangeReliefWidth

Relief width

FlangeReliefDepth

Relief depth

FlangeSharpType

Flange virtual sharp type as defined by [swFlangeDimTypes_e](#)

CustomBendAllowance

If...

Then...

non-NULL	Pointer to ICustomBendAllowance object for which required values have been set
NULL	Parent bend's bend allowance is used

Return Value

Pointer to the [IFeature](#) object

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertSheetMetalEdgeFlange2](#).

Remarks

Before calling this method, call [IModelDoc2.InsertSketchForEdgeFlange](#) and create a profile for the flange. After creating the profile, call this method to create the flange.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertSheetMetalEdgeFlange Method](#)
[EdgeFlangeFeatureData Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertTableDrivenPattern Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertTableDrivenPattern Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager.InsertTableDrivenPattern2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IInsertTableDrivenPattern( _
    ByVal FileName As System.String, _
    ByVal Count As System.Integer, _
    ByRef PointArr As System.Double, _
    ByVal UseCentrod As System.Boolean, _
    ByVal GeomPatt As System.Boolean _
) As Feature
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager
Dim FileName As System.String
Dim Count As System.Integer
Dim PointArr As System.Double
Dim UseCentrod As System.Boolean
Dim GeomPatt As System.Boolean
Dim value As Feature

value = instance.IInsertTableDrivenPattern(FileName, Count, PointArr, UseCentrod, GeomPatt)
```

C#

```
Feature IInsertTableDrivenPattern(
    System.string FileName,
    System.int Count,
    ref System.double PointArr,
    System.bool UseCentrod,
    System.bool GeomPatt
)
```

C++/CLI

```

Feature^ IInsertTableDrivenPattern(
& System.String^ FileName,
& System.int Count,
& System.double% PointArr,
& System.bool UseCentrod,
& System.bool GeomPatt
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Name of the file that has the coordinates information; can be an empty string (see **Remarks**)

Count

Number of points specified

PointArr

Array of x, y coordinate points (see **Remarks**)

UseCentrod

True to use the centroid of the seed feature, face, or body; false to use another point as the reference point (see **Remarks**)

GeomPatt

True to pattern the geometry, false to not

Return Value

[Feature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::IInsertTableDrivenPattern](#).

Remarks

You can specify:

- coordinates for the input points in system units in a **.sldptab** or **.txt** file for *FileName*.
- or -
- input points in system units in *PointArr*. Because each point has two coordinates (x, y), the size of *PointArr* is (2 x number_of_points).

This method requires selecting the input entities using these selection marks:

- 4 = Seed feature
- 16 = Coordinate system
- 32 = Reference point
- 128 = Seed face
- 256 = Seed body

If *UseCentrod* is false, then you must specify a reference point.

See the SOLIDWORKS Help for more information about table-driven patterns.

See Also

[IFeatureManager Interface](#)

[IFeatureManager Members](#)

[ITablePatternFeatureData Interface](#)

[IFeatureManager.InsertTableDrivenPattern Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCenterOfMass Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCenterOfMass Method (IFeatureManager)

Inserts a Center of Mass feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCenterOfMass() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As Feature

value = instance.InsertCenterOfMass()
```

C#

```
Feature InsertCenterOfMass()
```

C++/CLI

```
Feature^ InsertCenterOfMass();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCenterOfMass](#).

Example

[Insert Center of Mass Feature \(VBA\)](#)
[Insert Center of Mass Feature \(VB.NET\)](#)
[Insert Center of Mass Feature \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertCenterOfMassReferencePoint Method](#)
[ICenterOfMass Interface](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCenterOfMassReferencePoint Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCenterOfMassReferencePoint Method (IFeatureManager)

Inserts a Center of Mass Reference Point feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCenterOfMassReferencePoint() As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager
Dim value As Feature

value = instance.InsertCenterOfMassReferencePoint()
```

C#

```
Feature InsertCenterOfMassReferencePoint()
```

C++/CLI

```
Feature^ InsertCenterOfMassReferencePoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertCenterOfMassReferencePoint](#).

Example

[Insert Center of Mass Feature \(VBA\)](#)
[Insert Center of Mass Feature \(VB.NET\)](#)
[Insert Center of Mass Feature \(C#\)](#)

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager.InsertCenterOfMass Method](#)
[ICenterOfMass Interface](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCombineFeature Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertCombineFeature Method (IFeatureManager)

Combines the specified bodies in the multibody part to create a combine feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCombineFeature( _  
    ByVal OperationType As System.Integer, _  
    ByVal MainBody As Body2, _  
    ByVal ToolVar As System.Object _  
) As Feature
```

Visual Basic (Usage)



```
Dim instance As IFeatureManager  
Dim OperationType As System.Integer  
Dim MainBody As Body2  
Dim ToolVar As System.Object  
Dim value As Feature  
  
value = instance.InsertCombineFeature(OperationType, MainBody, ToolVar)
```

C#

```
Feature InsertCombineFeature(  
    System.int OperationType,  
    Body2 MainBody,  
    System.object ToolVar  
)
```

C++/CLI

```
Feature^ InsertCombineFeature(  
& System.int OperationType,  
& Body2^ MainBody,  
& System.Object^ ToolVar  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OperationType
Operation as defined in [swBodyOperationType_e](#)

MainBody

If OperationType is ...	Then set MainBody to ...
swBodyOperationType_e_SWBODYADD	Nothing or null
swBodyOperationType_e_SWBODYCUT	Target body
swBodyOperationType_e_SWBODYINTERSECT	Nothing or null

(See **Remarks**)
ToolVar
 Array of [bodies](#) to combine (see **Remarks**)

Return Value

[IFeature](#)

Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertCombineFeature](#).

Example

[Combine Bodies \(C#\)](#),
[Combine Bodies \(VB.NET\)](#),
[Combine Bodies \(VBA\)](#)

Remarks

You can either call this method, directly passing in the bodies with MainBody and ToolVar, or you can:

- Select the bodies using [IModelDocExtension::SelectByID2](#) as follows:

To select...	Use Mark...
MainBody	1
ToolVar bodies	2

- Call this method, setting MainBody to Nothing or null and ToolVar to an empty array.

See the SOLIDWORKS Help for more information about the combine feature.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IFeatureManager::InsertCombineFeature Method](#)
[ICombineBodiesFeatureData Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertConnectionPoint Method (IFeatureManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertConnectionPoint Method (IFeatureManager)

Adds a connection point based on the selected entities.

.NET Syntax

Visual Basic (Declaration)
<pre> Function InsertConnectionPoint(_ ByVal Type As System.Integer, _ ByVal SubType As System.Integer, _ ByVal ReverseDirection As System.Boolean, _ ByVal Diameter As System.Double, _ ByVal StubLength As System.Double, _ ByVal MinimumStraight As System.Double, _ ByVal EndLengthAdjustment As System.Double, _ ByVal ConductorSplitLength As System.Double, _ ByVal SchematicPinID As System.String, _ ByVal CableWidth As System.Double, _ ByVal CableThickness As System.Double, _ ByVal SwitchPin1Position As System.Boolean, _ ByVal SpecificationFieldName As System.String, _ ByVal SpecificationValue As System.String _) As System.Boolean </pre>

```

Dim instance As IFeatureManager
Dim Type As System.Integer
Dim SubType As System.Integer
Dim ReverseDirection As System.Boolean
Dim Diameter As System.Double
Dim StubLength As System.Double
Dim MinimumStraight As System.Double
Dim EndLengthAdjustment As System.Double
Dim ConductorSplitLength As System.Double
Dim SchematicPinID As System.String
Dim CableWidth As System.Double
Dim CableThickness As System.Double
Dim SwitchPin1Position As System.Boolean
Dim SpecificationFieldName As System.String
Dim SpecificationValue As System.String
Dim value As System.Boolean

value = instance.InsertConnectionPoint(Type, SubType, ReverseDirection, Diameter, StubLength,
MinimumStraight, EndLengthAdjustment, ConductorSplitLength, SchematicPinID, CableWidth, CableThickness,
SwitchPin1Position, SpecificationFieldName, SpecificationValue)

```

C#

```

System.bool InsertConnectionPoint(
    System.int Type,
    System.int SubType,
    System.bool ReverseDirection,
    System.double Diameter,
    System.double StubLength,
    System.double MinimumStraight,
    System.double EndLengthAdjustment,
    System.double ConductorSplitLength,
    System.string SchematicPinID,
    System.double CableWidth,
    System.double CableThickness,
    System.bool SwitchPin1Position,
    System.string SpecificationFieldName,
    System.string SpecificationValue
)

```

C++/CLI

```

System.bool InsertConnectionPoint(
    & System.int Type,
    & System.int SubType,
    & System.bool ReverseDirection,
    & System.double Diameter,
    & System.double StubLength,
    & System.double MinimumStraight,
    & System.double EndLengthAdjustment,
    & System.double ConductorSplitLength,
    & System.String^ SchematicPinID,
    & System.double CableWidth,
    & System.double CableThickness,
    & System.bool SwitchPin1Position,
    & System.String^ SpecificationFieldName,
    & System.String^ SpecificationValue
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Route type as defined in [swConnectionPointType_e](#)

SubType

Electrical route sub-type as defined in [swElectricalConnectionPointType_e](#)

ReverseDirection

True to flip the direction in which the route exits the fitting, false to not

Diameter

Nominal diameter (for pipe, tube, and electrical conduit fittings only)

StubLength

Default stub length to extend from the connector or fitting when inserted into routes

NOTE: For ribbon cables, leave stub length at 0 or set it to a value greater than half the width of the cable to assure that folds made near the connector can be created properly.

MinimumStraight
Minimum straight length (for tubes only)
EndLengthAdjustment
End length adjustment (for tubes only)
ConductorSplitLength
Conductor split length (for electrical only)
SchematicPinID
2D schematic pin ID (for ribbon cables only)
CableWidth
Width (for ribbon cables only)
CableThickness
Cable thickness (for ribbon cables only)
SwitchPin1Position
Switch Pin1 position (for ribbon cables only)
SpecificationFieldName
Specification field name (for pipes and tubes only)
SpecificationValue
Specification value (for pipes and tubes only)

Return Value

True if the connection is inserted, false if not

 Visual Basic for Applications (VBA) Syntax

See [FeatureManager::InsertConnectionPoint](#).

 Example

[Insert Connection Point \(C#\)](#)
[Insert Connection Point \(VB.NET\)](#)
[Insert Connection Point \(VBA\)](#)

 Remarks

This method only works when the SOLIDWORKS Routing add-in is loaded (click **Tools > Add-Ins** and select **SOLIDWORKS Routing**). This method currently supports part documents only.

Use [IModelDocExtension::SelectByID2](#) with:

- Mark = 1, which can be any one of the following entities to indicate the origin of a route:
 - Circular edge
 - Circular face
 - Face or plane plus a sketch point or vertex (select the plane or face where the routing segment exits the fitting, then select the sketch point that defines the end of the adjoining route segment)
- Mark = 2, which can be a reference axis or edge to indicate which entity to align the width of the cable (for ribbon cables only)

See the SOLIDWORKS Help for details about connection points and routing.

 See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)
[IAssemblyDoc::GetRouteManager Method](#)
[IModelDoc2::InsertRoutePoint Method](#)
[IConnectionPointFeatureData Interface](#)

 Availability

SOLIDWORKS 2009 SP03, Revision Number 17.03

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertConvertToSheetMetal Method (IFeatureManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureManager Interface](#) : InsertConvertToSheetMetal Method (IFeatureManager)

Obsolete. Superseded by [IFeatureManager::InsertConvertToSheetMetal2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function InsertConvertToSheetMetal( _  
    ByVal Thickness As System.Double, _  
    ByVal ReverseThickDir As System.Boolean, _  
    ByVal FindBends As System.Boolean, _  
    ByVal Radius As System.Double, _  
    ByVal Gap As System.Double, _  
    ByVal ReliefType As System.Integer, _  
    ByVal ReliefRatio As System.Double _  
) As System.Boolean
```



```

Dim instance As IFeatureManager
Dim Thickness As System.Double
Dim ReverseThickDir As System.Boolean
Dim FindBends As System.Boolean
Dim Radius As System.Double
Dim Gap As System.Double
Dim ReliefType As System.Integer
Dim ReliefRatio As System.Double
Dim value As System.Boolean

value = instance.InsertConvertToSheetMetal(Thickness, ReverseThickDir, FindBends, Radius, Gap,
ReliefType, ReliefRatio)

```

C#

```

System.bool InsertConvertToSheetMetal(
    System.double Thickness,
    System.bool ReverseThickDir,
    System.bool FindBends,
    System.double Radius,
    System.double Gap,
    System.int ReliefType,
    System.double ReliefRatio
)

```

C++/CLI

```

System.bool InsertConvertToSheetMetal(
& System.double Thickness,
& System.bool ReverseThickDir,
& System.bool FindBends,
& System.double Radius,
& System.double Gap,
& System.int ReliefType,
& System.double ReliefRatio
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness
Sheet thickness
ReverseThickDir
True to reverse the direction of the sheet thickness, false to not
FindBends
True to find pre-made bends and part thickness, false to not
Radius
Radius for the bends
Gap
Gap for all rips
ReliefType
Relief type as defined by [swSheetMetalReliefTypes_e](#)
ReliefRatio
Relief ratio

Return Value

True if a Convert-Solid sheet metal feature is created, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertConvertToSheetMetal](#).

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SolidWorks 2009 FCS, Revision Number 17.0



.NET Syntax

Visual Basic (Declaration)

```
Function InsertConvertToSheetMetal2( _  
    ByVal Thickness As System.Double, _  
    ByVal ReverseThickDir As System.Boolean, _  
    ByVal FindBends As System.Boolean, _  
    ByVal Radius As System.Double, _  
    ByVal Gap As System.Double, _  
    ByVal ReliefType As System.Integer, _  
    ByVal ReliefRatio As System.Double, _  
    ByVal OverlapType As System.Integer, _  
    ByVal OverlapRatio As System.Double, _  
    ByVal KeepBody As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IFeatureManager  
Dim Thickness As System.Double  
Dim ReverseThickDir As System.Boolean  
Dim FindBends As System.Boolean  
Dim Radius As System.Double  
Dim Gap As System.Double  
Dim ReliefType As System.Integer  
Dim ReliefRatio As System.Double  
Dim OverlapType As System.Integer  
Dim OverlapRatio As System.Double  
Dim KeepBody As System.Boolean  
Dim value As System.Boolean  
  
value = instance.InsertConvertToSheetMetal2(Thickness, ReverseThickDir, FindBends, Radius, Gap,  
ReliefType, ReliefRatio, OverlapType, OverlapRatio, KeepBody)
```

C#

```
System.bool InsertConvertToSheetMetal2(  
    System.double Thickness,  
    System.bool ReverseThickDir,  
    System.bool FindBends,  
    System.double Radius,  
    System.double Gap,  
    System.int ReliefType,  
    System.double ReliefRatio,  
    System.int OverlapType,  
    System.double OverlapRatio,  
    System.bool KeepBody  
)
```

C++/CLI

```
System.bool InsertConvertToSheetMetal2(  
& System.double Thickness,  
& System.bool ReverseThickDir,  
& System.bool FindBends,  
& System.double Radius,  
& System.double Gap,  
& System.int ReliefType,  
& System.double ReliefRatio,  
& System.int OverlapType,  
& System.double OverlapRatio,  
& System.bool KeepBody  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Thickness
Sheet thickness
ReverseThickDir
True to reverse the direction of the sheet thickness, false to not
FindBends
True to find pre-made bends and part thickness, false to not
Radius
Radius for the bends
Gap
Gap for all rips
ReliefType
Relief type as defined by [swSheetMetalReliefTypes_e](#)
ReliefRatio
Relief ratio
OverlapType
Overlap type for all rips

- 1=Open butt
- 2=Overlap
- 3=Underlap

OverlapRatio
Overlap ratio for all rips
KeepBody
True to keep bodies, false to not

Return Value

True if a Convert-Solid sheet metal feature is created, false if not

Visual Basic for Applications (VBA) Syntax

See [FeatureManager.InsertConvertToSheetMetal2](#).

Remarks

Read the SOLIDWORKS Help to learn more about converting to sheet metal.

See Also

[IFeatureManager Interface](#)
[IFeatureManager Members](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0